



GROUP PROJECTS

TCN2141 – COMPUTER NETWORKS
TRIMESTER 1, 2023/2024 (TERM 2310)

Lecture Section: CN2

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Table of Contents

PART A: ENTREPRENEURIAL SECTION	2
1. PROJECT IMPLEMENTATION	2
1.1. GANTT CHART	2
1.2. PROJECT BUDGET	3
PART B: DESIGN, SETUP AND CONFIGURATION	5
2. NETWORK TOPOLOGY	5
2.1. LOGICAL TOPOLOGY	5
2.2. PHYSICAL TOPOLOGY	6
3. LOGICAL DESIGN	7
3.1. ADDRESSING TABLE	7
4. ROUTER CONFIGURATION	15
5. IPV6 ROUTER CONFIGURATION	23
6. DHCP SERVER CONFIGURATION	27
7. OSPF CONFIGURATION	31
8. CLOUD AND ISP ROUTER CONFIGURATION	34
9. ACCESS CONTROL LIST	36
9.1. DENY INTERNET ACCESS TO ADMIN AND TUTOR ROOM	36
9.2. DENY INTERNET ACCESS TO WIRELESS NETWORK	37
10. WEB SERVER CONFIGURATION	38
11. FTP Server Configuration	40
12. DNS SERVER CONFIGURATION	44
13. TM SERVER CONFIGURATION	46
14. CONNECTIVITY STATUS	49

PART A: ENTREPRENEURIAL SECTION

1. PROJECT IMPLEMENTATION

1.1.GANTT CHART

Task	October				November				December				January			
	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4
1.0 Preparation																
Task distribution																
2.0 Planning																
Identifying Users																
Possible Hardware																
Type of Networks																
Topology Arrangement																
3.0 Design																
Network Analysis																
Logical Design																
Physical Design																
Topology Mockup																
4.0 Configuration																
Configuration Planning																
Network Setup																
Testing and Troubleshooting																
5.0 Report																
Configuration Documentation																
Report Finalization																
Video demo																
Submission																

1.2.PROJECT BUDGET

COST ANALYSIS

A. NETWROK HARDWARE AND COMPONENT			
Product	Price (RM)	QTY	Total (RM)
S3910-24TF, 24-Port Gigabit Ethernet L2+ Switch	RM1,709.00	7	RM11,963.00
Cisco 2901 Router ISR G2 CISCO2901/K9	RM2,620.00	1	RM2,620.00
Cisco 2911 Integrated Services Router (2911 Router)	RM1,000.00	4	RM4,000.00
Student Workstation	RM3,500.00	48	RM168,000.00
Administrator/ Tutor/Instructor Workstation	RM5,000.00	24	RM120,000.00
PowerEdge R7515 Rack Server	RM18,910.14	1	RM18,910.14
Refurbished Cisco Access Point AIR-CAP3702i-C-K9	RM1,900.00	1	RM1,900.00
GrowV 19' Floor Stand Rack Server Rack 24U (Perforated) (P/G2480FS)	RM1,250.00	3	RM3,750.00
FortiGate-80E Hardware plus 1 Year 24x7 FortiCare and FortiGuard Unified Threat Protection (UTP)	RM7,000.00	1	RM7,000.00
HP LaserJet P3015 (CE528A)	RM1,097.00	3	RM3,291.00
Seagate Skyhawk AI 16TB 3.5" HDD Surveillance NVR Hard Drive SATA 7200RPM Internal Hard Disk	RM1,165.00	7	RM8,155.00
Subtotal			RM349,589.14

B. Network software and Configuration			
Product	Price (RM)	Qty	Total (RM)
Microsoft Windows Server 2019 Standard OEM PACK Standard 64- Bit License (16 Core, OEM) (2 CPUs / 2VM) (P73- 07788)	RM3,500.00	1	RM 3,500.00
Microsoft Windows 11 License	RM 344.00	72	RM24,768.00
PRTG Network Monitor	RM -	1	RM -
Ubuntu 18.04 LTS	RM -	1	RM -
Vmware Workstation 17.5	RM 42.77	1	RM 42.77

Subtotal	RM28,310.77
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C. Cabling and Sockets components			
Product	Price (RM)	Qty	Total (RM)
CAT-6 Ethernet Cable (30M)	RM 45.00	7	RM 315.00
RJ45 Cat6 Network Plugs/ Connector	RM 50.00	70	RM 3,500.00
RS232 Serial DB9 9Pin Male to Male Extension Cable (5M)	RM 30.00	6	RM 180.00
Subtotal			RM 3,995.00

D. Man-hour requirement			
Task	Rate per Hour (RM)	Qty	Total (RM)
Project Manager	RM 30.00	320	RM 9,600.00
Network Engineer	RM 40.00	320	RM12,800.00
Technician	RM 15.00	250	RM 3,750.00
Subtotal			RM26,150.00

Table 1 Cost Analysis table

PART B: DESIGN, SETUP AND CONFIGURATION

2. NETWORK TOPOLOGY

2.1.LOGICAL TOPOLOGY

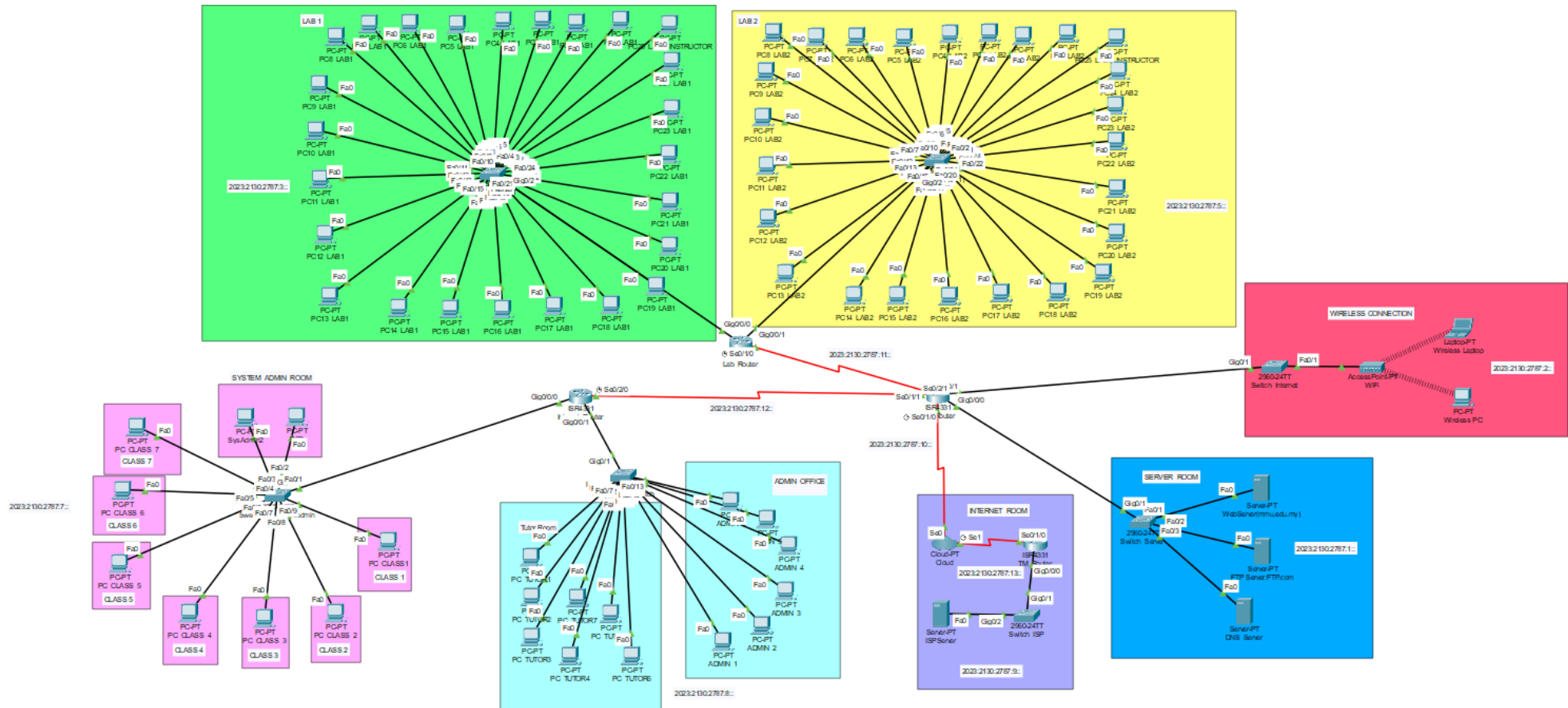


Figure 1- Logical Topology

2.2. PHYSICAL TOPOLOGY

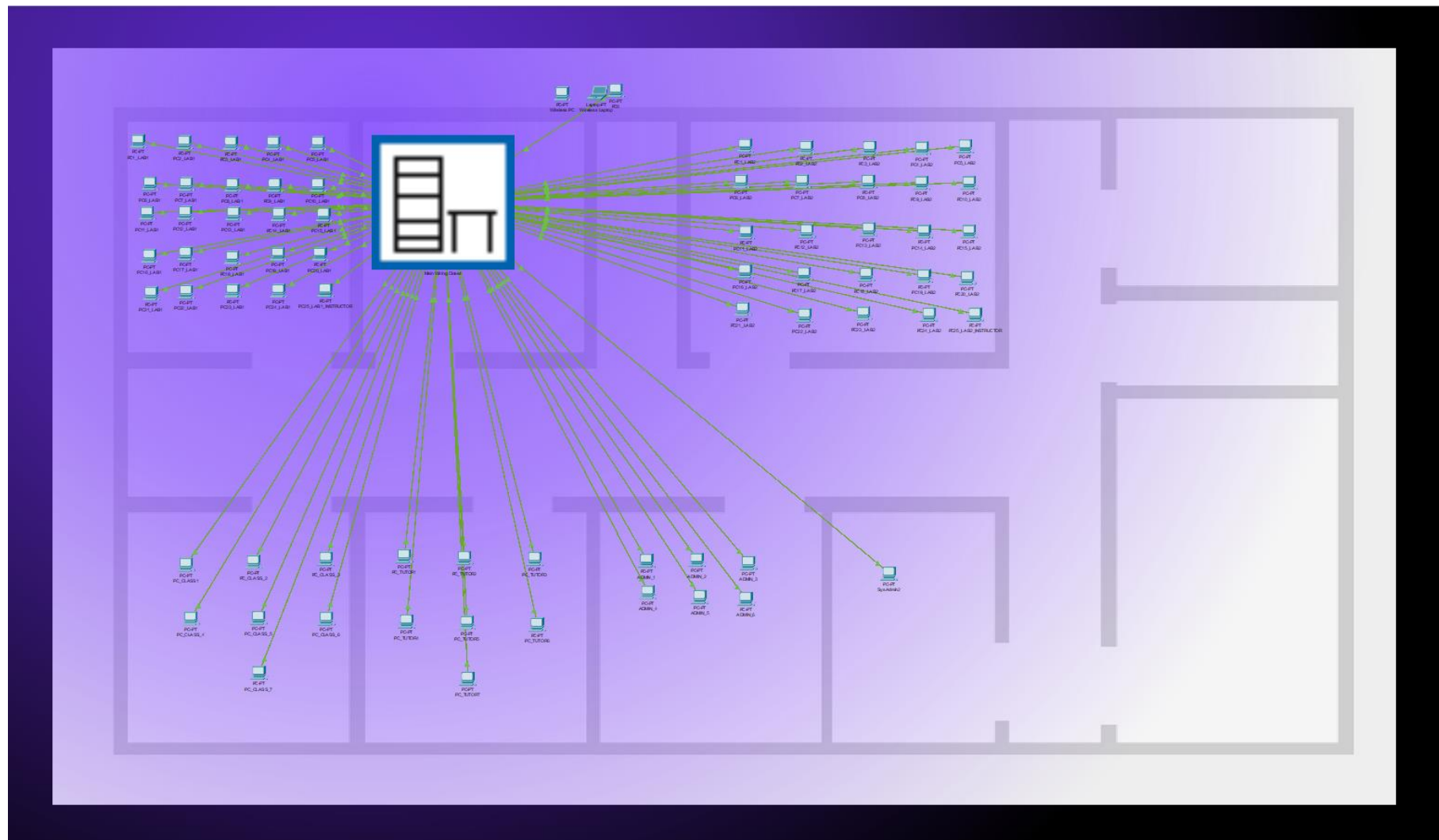


Figure 2 Physical Topology

3. LOGICAL DESIGN

3.1.ADDRESSING TABLE

LAB 1 IPV6 ADDRESS				
PC1_LAB1	FastEthernet 0	SWITCH_LAB1 FastEthernet 0/1	2023:2130:2787:3:BD15:91E1:835F:835F	FE80::3:1
PC2_LAB1	FastEthernet 0	SWITCH_LAB1 FastEthernet 0/2	2023:2130:2787:3:9166:7632:68A0:5A1E	FE80::3:1
PC3_LAB1	FastEthernet 0	SWITCH_LAB1 FastEthernet 0/3	2023:2130:2787:3:878B:79E8:6B56:5DB4	FE80::3:1
PC4_LAB1	FastEthernet 0	SWITCH_LAB1 FastEthernet 0/4	2023:2130:2787:3:E03F:D29D:C40B:B669	FE80::3:1
PC5_LAB1	FastEthernet 0	SWITCH_LAB1 FastEthernet 0/5	2023:2130:2787:3:998C:8BFA:7D58:6FC5	FE80::3:1
PC6_LAB1	FastEthernet 0	SWITCH_LAB1 FastEthernet 0/6	2023:2130:2787:3:F729:A23E:94AC:860A	FE80::3:1
PC7_LAB1	FastEthernet 0	SWITCH_LAB1 FastEthernet 0/7	2023:2130:2787:3:91FD:6736:5994:4B02	FE80::3:1
PC8_LAB1	FastEthernet 0	SWITCH_LAB1 FastEthernet 0/8	2023:2130:2787:3:51CF:35AB:2709:2709	FE80::3:1
PC9_LAB1	FastEthernet 0	SWITCH_LAB1 FastEthernet 0/9	2023:2130:2787:3:A87C:9AEA:7FB6:7FB6	FE80::3:1
PC10_LAB1	FastEthernet 0	SWITCH_LAB1 FastEthernet 0/10	2023:2130:2787:3:E71D:D97B:BD57:BD57	FE80::3:1
PC11_LAB1	FastEthernet 0	SWITCH_LAB1 FastEthernet 0/11	2023:2130:2787:3:5411:38ED:2A5B:2A5B	FE80::3:1

PC12_LAB1	FastEthernet 0	SWITCH_LAB1 FastEthernet 0/12	2023:2130:2787:3:A699:9807:7DD3:7DD3	FE80::3:1
PC13_LAB1	FastEthernet 0	SWITCH_LAB1 FastEthernet 0/13	2023:2130:2787:3:373B:29A9:1B07:E75	FE80::3:1
PC14_LAB1	FastEthernet 0	SWITCH_LAB1 FastEthernet 0/14	2023:2130:2787:3:6EFA:42D6:3433:3433	FE80::3:1
PC15_LAB1	FastEthernet 0	SWITCH_LAB1 FastEthernet 0/15	2023:2130:2787:3:E442:ACEA:9E58:80B6	FE80::3:1
PC16_LAB1	FastEthernet 0	SWITCH_LAB1 FastEthernet 0/16	2023:2130:2787:3:A23D:949B:8609:7966	FE80::3:1
PC17_LAB1	FastEthernet 0	SWITCH_LAB1 FastEthernet 0/17	2023:2130:2787:3:68E:EB6A:DDC8:CF36	FE80::3:1
PC18_LAB1	FastEthernet 0	SWITCH_LAB1 FastEthernet 0/18	2023:2130:2787:3:EFE9:C3B4:B512:B512	FE80::3:1
PC19_LAB1	FastEthernet 0	SWITCH_LAB1 FastEthernet 0/19	2023:2130:2787:3:386D:1C38:EA6:F104	FE80::3:1
PC20_LAB1	FastEthernet 0	SWITCH_LAB1 FastEthernet 0/20	2023:2130:2787:3:5B2D:3FF9:2157:13C5	FE80::3:1
PC21_LAB1	FastEthernet 0	SWITCH_LAB1 FastEthernet 0/21	2023:2130:2787:3:686:EA52:DCC0:C E2E	FE80::3:1
PC22_LAB1	FastEthernet 0	SWITCH_LAB1 FastEthernet 0/22	2023:2130:2787:3:E14D:D3AB:C519:B787	FE80::3:1
PC23_LAB1	FastEthernet 0	SWITCH_LAB1 FastEthernet 0/23	2023:2130:2787:3:B8A:C332:B590:B590	FE80::3:1
PC24_LAB1	FastEthernet 0	SWITCH_LAB1 FastEthernet 0/24	2023:2130:2787:3:89D3:7B31:6DAF:5F0C	FE80::3:1
PC25_LAB1_INSTRUCTOR	FastEthernet 0	SWITCH_LAB1 GigabitEthernet 0/1	2023:2130:2787:3:D6F1:AC3B:8107:8107	FE80::3:1

LAB 2 IPV6 ADDRESS				
PC1_LAB2	FastEthernet 0	SWITCH_LAB2 FastEthernet 0/1	2023:2130:2787:5:87A3:7901:5DDC:5 DDC	FE80::4:1
PC2_LAB2	FastEthernet 0	SWITCH_LAB2 FastEthernet 0/2	2023:2130:2787:5:F203:D6DF:D6DF:C 83D	FE80::4:1
PC3_LAB2	FastEthernet 0	SWITCH_LAB2 FastEthernet 0/3	2023:2130:2787:5:58CE:1076:2D3:F44 1	FE80::4:1
PC4_LAB2	FastEthernet 0	SWITCH_LAB2 FastEthernet 0/4	2023:2130:2787:5:5548:1DEF:9FDD:8 13A	FE80::4:1
PC5_LAB2	FastEthernet 0	SWITCH_LAB2 FastEthernet 0/5	2023:2130:2787:5:65A6:5814:4A71:3C EF	FE80::4:1
PC6_LAB2	FastEthernet 0	SWITCH_LAB2 FastEthernet 0/6	2023:2130:2787:5:91AF:757B:67E9:59 56	FE80::4:1
PC7_LAB2	FastEthernet 0	SWITCH_LAB2 FastEthernet 0/7	2023:2130:2787:5:9907:8B65:7ED3:60 31	FE80::4:1
PC8_LAB2	FastEthernet 0	SWITCH_LAB2 FastEthernet 0/8	2023:2130:2787:5:68A:F8E8:EA55:DC B3	FE80::4:1
PC9_LAB2	FastEthernet 0	SWITCH_LAB2 FastEthernet 0/9	2023:2130:2787:5:B530:9A0B:8C69:7E D7	FE80::4:1
PC10_LAB2	FastEthernet 0	SWITCH_LAB2 FastEthernet 0/10	2023:2130:2787:5:595C:4BBA:3D27:3 D27	FE80::4:1
PC11_LAB2	FastEthernet 0	SWITCH_LAB2 FastEthernet 0/11	2023:2130:2787:5:8CA9:52E3:4441:37 BF	FE80::4:1
PC12_LAB2	FastEthernet 0	SWITCH_LAB2 FastEthernet 0/12	2023:2130:2787:5:A603:8AEE:7C4C:7 C4C	FE80::4:1
PC13_LAB2	FastEthernet 0	SWITCH_LAB2 FastEthernet 0/13	2023:2130:2787:5:5050:42CE:342C:26 99	FE80::4:1
PC14_LAB2	FastEthernet 0	SWITCH_LAB2 FastEthernet 0/14	2023:2130:2787:5:CF76:A342:95B0:95 B0	FE80::4:1

PC15_LAB2	FastEthernet 0	SWITCH_LAB2 FastEthernet 0/15	2023:2130:2787:5:9341:5BF8:4D56:3F C4	FE80::4:1
PC16_LAB2	FastEthernet 0	SWITCH_LAB2 FastEthernet 0/16	2023:2130:2787:5:F610:DAEC:DAEC: CC5A	FE80::4:1
PC17_LAB2	FastEthernet 0	SWITCH_LAB2 FastEthernet 0/17	2023:2130:2787:5:EE61:D0DE:C33C:C 33C	FE80::4:1
PC18_LAB2	FastEthernet 0	SWITCH_LAB2 FastEthernet 0/18	2023:2130:2787:5:77E3:6951:5BBF:4D 2C	FE80::4:1
PC19_LAB2	FastEthernet 0	SWITCH_LAB2 FastEthernet 0/19	2023:2130:2787:5:A616:9874:8AE2:7C 50	FE80::4:1
PC20_LAB2	FastEthernet 0	SWITCH_LAB2 FastEthernet 0/20	2023:2130:2787:5:6F8:EAC4:CEAF:C EAF	FE80::4:1
PC21_LAB2	FastEthernet 0	SWITCH_LAB2 FastEthernet 0/21	2023:2130:2787:5:C61D:AAE8:9C56:8 EC4	FE80::4:1
PC22_LAB2	FastEthernet 0	SWITCH_LAB2 FastEthernet 0/22	2023:2130:2787:5:624B:4617:3885:388 5	FE80::4:1
PC23_LAB2	FastEthernet 0	SWITCH_LAB2 FastEthernet 0/23	2023:2130:2787:5:9901:8B7F:7DDD:7 DDD	FE80::4:1
PC24_LAB2	FastEthernet 0	SWITCH_LAB2 FastEthernet 0/24	2023:2130:2787:5:E0F7:D265:C4D3:B 631	FE80::4:1
PC25_LAB2_INSTRUCT OR	FastEthernet 0	SWITCH_LAB2 GigabitEthernet 0/1	2023:2130:2787:5:128F:DA27:CC95:B EF2	FE80::4:1

TUTOR ROOM IPV6 ADDRESS				
PC_TUTOR1	FastEthernet 0	SWITCH_TUTOR_AD MIN	2023:2130:2787:8:3FC9:6F2:EADE:EA DE	FE80::9:1

		FastEthernet 0/1		
PC_TUTOR2	FastEthernet 0	SWITCH_TUTOR_AD MIN FastEthernet 0/2	2023:2130:2787:8:188D:AEB:FC59:EE C7	FE80::9:1
PC_TUTOR3	FastEthernet 0	SWITCH_TUTOR_AD MIN FastEthernet 0/3	2023:2130:2787:8:2F2D:E8C4:DA32:C C90	FE80::9:1
PC_TUTOR4	FastEthernet 0	SWITCH_TUTOR_AD MIN FastEthernet 0/4	2023:2130:2787:8:D60C:BAE8:AC46:9 EB4	FE80::9:1
PC_TUTOR5	FastEthernet 0	SWITCH_TUTOR_AD MIN FastEthernet 0/5	2023:2130:2787:8:329F:166B:8D9:FA3 7	FE80::9:1
PC_TUTOR6	FastEthernet 0	SWITCH_TUTOR_AD MIN FastEthernet 0/6	2023:2130:2787:8:A78:FCD6:EE44:D0 B2	FE80::9:1
PC_TUTOR7	FastEthernet 0	SWITCH_TUTOR_AD MIN FastEthernet 0/7	2023:2130:2787:8:470:F6DE:E84C:DA A9	FE80::9:1
ADMIN OFFICE IPV6 ADDRESS				
ADMIN_1	FastEthernet 0	SWITCH_TUTOR_AD MIN FastEthernet 0/8	2023:2130:2787:5:68A:F8E8:EA55:DC B3	FE80::4:1
ADMIN_2	FastEthernet 0	SWITCH_TUTOR_AD MIN FastEthernet 0/9	2023:2130:2787:5:B530:9A0B:8C69:7E D7	FE80::4:1
ADMIN_3	FastEthernet 0	SWITCH_TUTOR_AD MIN FastEthernet 0/10	2023:2130:2787:5:595C:4BBA:3D27:3 D27	FE80::4:1

ADMIN_4	FastEthernet 0	SWITCH_TUTOR_AD MIN FastEthernet 0/11	2023:2130:2787:5:8CA9:52E3:4441:37 BF	FE80::4:1
ADMIN_5	FastEthernet 0	SWITCH_TUTOR_AD MIN FastEthernet 0/12	2023:2130:2787:5:A603:8AEE:7C4C:7 C4C	FE80::4:1
ADMIN_6	FastEthernet 0	SWITCH_TUTOR_AD MIN FastEthernet 0/13	2023:2130:2787:5:5050:42CE:342C:269 9	FE80::4:1
ADMIN_7	FastEthernet 0	SWITCH_TUTOR_AD MIN FastEthernet 0/14	2023:2130:2787:5:CF76:A342:95B0:95 B0	FE80::4:1

SYSTEM ADMIN ROOM IPV6 ADDRESS				
SYSADMIN1	FastEthernet 0	SWITCH_CLASS_SYS ADMIN FastEthernet 0/1	2023:2130:2787:7:FA60:DE3C:C0AA:C 0AA	FE80::8:1
SYSADMIN2	FastEthernet 0	SWITCH_CLASS_SYS ADMIN FastEthernet 0/2	2023:2130:2787:7:BF38:A1A6:9314:857 2	FE80::8:1
CLASS IPV6 ADDRESS				
CLASS_1	FastEthernet 0	SWITCH_CLASS_SYS ADMIN FastEthernet 0/9	2023:2130:2787:7:1F5B:F327:E594:E59 4	FE80::8:1

CLASS_2	FastEthernet 0	SWITCH_CLASS_SYS ADMIN FastEthernet 0/8	2023:2130:2787:7:3588:1954:BC2:FD20	FE80::8:1
CLASS_3	FastEthernet 0	SWITCH_CLASS_SYS ADMIN FastEthernet 0/7	2023:2130:2787:7:6077:53D5:4542:4542	FE80::8:1
CLASS_4	FastEthernet 0	SWITCH_CLASS_SYS ADMIN FastEthernet 0/6	2023:2130:2787:7:5101:437E:35EC:274 A	FE80::8:1
CLASS_5	FastEthernet 0	SWITCH_CLASS_SYS ADMIN FastEthernet 0/5	2023:2130:2787:7:472D:2BF9:C86C:BA DA	FE80::8:1
CLASS_6	FastEthernet 0	SWITCH_CLASS_SYS ADMIN FastEthernet 0/4	2023:2130:2787:7:60C7:44A3:3701:370 1	FE80::8:1
CLASS_7	FastEthernet 0	SWITCH_CLASS_SYS ADMIN FastEthernet 0/3	2023:2130:2787:7:CD55:939F:776B:776 B	FE80::8:1
WIRELESS IPV6 ADDRESS				
WIRELESS PC	Wireless	SWITCH_INTERNET FastEthernet 0/1	2023:2130:2787:2:D40B:B9E6:AB44:9D B2	FE80::2:1
WIRELESS LAPTOP	Wireless	SWITCH_INTERNET FastEthernet 0/2	2023:2130:2787:2:89AE:4245:2621:2621	FE80::2:1

DEVICE	CONNECTED FROM	CONNECTED TO	IP ADDRESS	LINK-LOCAL
MAIN_ROUTER	GigabitEthernet 0/0/0	Switch_Server GigabitEthernet 0/2	2023:2130:2787:1::1	FE80::1:1
	GigabitEthernet 0/0/1	Switch_Internet	2023:2130:2787:2::1	FE80::2:1

		GigabitEthernet 0/1		
	Serial 0/1/0	Cloud-PT Cloud Serial 0	2023:2130:2787:10::1	FE80::5:1
	Serial 0/1/1	Internal_Router Serial 0/2/0	2023:2130:2787:11::1	FE80::5:2
	Serial 0/2/1	Lab_Router Serial 0/1/0	2023:2130:2787:12::1	FE80::5:3
LAB_ROUTER	GigabitEthernet 0/0/0	Switch_LAB1 GigabitEthernet 0/2	2023:2130:2787:3::1	FE80::3:1
	GigabitEthernet 0/0/1	Switch_LAB2 GigabitEthernet 0/2	2023:2130:2787:5::1	FE80::4:1
	Serial0/1/0	Main_Router Serial 0/2/0	2023:2130:2787:11::2	FE80::5:4
INTERNAL_ROUTER	GigabitEthernet 0/0/0	Switch_Class_Sysadmin GigabitEthernet 0/1	2023:2130:2787:7::1	FE80::8:1
	GigabitEthernet 0/0/1	Tutor_Admin_Switch GigabitEthernet 0/1	2023:2130:2787:8::1	FE80::9:1
	Serial0/2/0	Main_Router Serial 0/1/1	2023:2130:2787:12::2	FE80::5:6
TM_ROUTER	GigabitEthernet 0/0/0	Switch_ISP GigabitEthernet 0/1	2023:2130:2787:9::1	FE80::10:1
	Serial 0/1/0	Cloud-PT Cloud Serial 1	2023:2130:2787:13::1	FE80::5:7
FTP_SERVER	FastEthernet 0	Switch_Server FastEthernet 0/1	2023:2130:2787:1::5	2023:2130:2787:1::1
WEB_SERVER	FastEthernet 0	Switch_Server FastEthernet 0/2	2023:2130:2787:1::4	2023:2130:2787:1::1
DNS_SERVER	FastEthernet 0	Switch_Server FastEthernet 0/3	2023:2130:2787:1::300	2023:2130:2787:1::1
ISP_SERVER	FastEthernet 0	Switch_Server GigabitEthernet 0/2	2023:2130:2787:9::5	2023:2130:2787:9::1

4. ROUTER CONFIGURATION

Main Router

```
Router>enable
```

```
Router#configure terminal
```

```
Router(config)#hostname Main_Router
```

```
Main_Router(config)#ipv6 unicast-routing
```

```
Main_Router(config)#ipv6 dhcp pool STATEFUL_1
```

```
Main_Router(config-dhcpv6)#address prefix 2023:2130:2787:1::/64
```

```
Main_Router(config-dhcpv6)#dns-server 2023:2130:2787:1::300
```

```
Main_Router(config-dhcpv6)#domain-name Main_server.com
```

```
Main_Router(config-dhcpv6)#exit
```

```
Main_Router(config)#ipv6 dhcp pool STATEFUL_2
```

```
Main_Router(config-dhcpv6)#address prefix 2023:2130:2787:2::/64
```

```
Main_Router(config-dhcpv6)#dns-server 2023:2130:2787:1::300
```

```
Main_Router(config-dhcpv6)#domain-name Main_wifi.com
```

```
Main_Router(config-dhcpv6)#exit
```

```
Main_Router(config)#ipv6 router ospf 10
```

```
Main_Router(config-rtr)#router-id 1.1.1.1
```

```
Main_Router(config-rtr)#exit
```

```
Main_Router(config)#interface GigabitEthernet 0/0/0
```

```
Main_Router(config-if)#ipv6 address 2023:2130:2787:1::1/64
```

```
Main_Router(config-if)#ipv6 address FE80::1:1 link-local
Main_Router(config-if)#ipv6 nd managed-config-flag
Main_Router(config-if)#ipv6 dhcp server STATEFUL_1
Main_Router(config-if)#ipv6 ospf 10 area 0
Main_Router(config-if)#no shutdown
Main_Router(config-if)#exit
```

```
Main_Router(config)#interface GigabitEthernet 0/0/1
Main_Router(config-if)#ipv6 address 2023:2130:2787:2::1/64
Main_Router(config-if)#ipv6 address FE80::2:1 link-local
Main_Router(config-if)#ipv6 nd managed-config-flag
Main_Router(config-if)#ipv6 dhcp server STATEFUL_2
Main_Router(config-if)#ipv6 ospf 10 area 0
Main_Router(config-if)#no shutdown
Main_Router(config-if)#exit
```

```
Main_Router(config)#interface Serial 0/1/0
Main_Router(config-if)#ipv6 address 2023:2130:2787:10::1/64
Main_Router(config-if)#ipv6 address FE80::5:1 link-local
Main_Router(config-if)#ipv6 enable
Main_Router(config-if)#ipv6 ospf 10 area 0
Main_Router(config-if)#no shutdown
Main_Router(config-if)#exit
```

```
Main_Router(config)#interface serial 0/1/1
Main_Router(config-if)#ipv6 address 2023:2130:2787:11::1/64
Main_Router(config-if)#ipv6 address FE80::5:2 link-local
Main_Router(config-if)#ipv6 enable
Main_Router(config-if)#ipv6 ospf 10 area 0
Main_Router(config-if)#no shutdown
Main_Router(config-if)#exit
```

```
Main_Router(config)#interface Serial 0/2/1
Main_Router(config-if)#ipv6 address 2023:2130:2787:12::1/64
Main_Router(config-if)#ipv6 address FE80::5:3 link-local
Main_Router(config-if)#ipv6 enable
Main_Router(config-if)#ipv6 ospf 10 area 0
Main_Router(config-if)#no shutdown
Main_Router(config-if)#end
```

```
Main_Router#copy run start
```

Lab_Router

```
Router>enable
```

```
Router#configure terminal
```

```
Router(config)#hostname Lab_Router
```

```
Lab_Router(config)#ipv6 unicast-routing
```

```
Lab_Router(config)#ipv6 dhcp pool STATEFUL_3
```

```
Lab_Router(config-dhcpv6)#address prefix 2023:2130:2787:3::/64
```

```
Lab_Router(config-dhcpv6)#dns-server 2023:2130:2787:1::300
```

```
Lab_Router(config-dhcpv6)#domain-name Lab_1.com
```

```
Lab_Router(config-dhcpv6)#exit
```

```
Lab_Router(config)#ipv6 dhcp pool STATEFUL_4
```

```
Lab_Router(config-dhcpv6)#address prefix 2023:2130:2787:5::/64
```

```
Lab_Router(config-dhcpv6)#dns-server 2023:2130:2787:1::300
```

```
Lab_Router(config-dhcpv6)#domain-name Lab_2.com
```

```
Lab_Router(config-dhcpv6)#exit
```

```
Lab_Router(config)#ipv6 router ospf 10
```

```
Lab_Router(config-rtr)#router-id 2.2.2.2
```

```
Lab_Router(config-rtr)#exit
```

```
Lab_Router(config)#interface GigabitEthernet 0/0/0
```

```
Lab_Router(config-if)#ipv6 address 2023:2130:2787:3::1/64
```

```
Lab_Router(config-if)#ipv6 address FE80::3:1 link-local
```

```
Lab_Router(config-if)#ipv6 nd managed-config-flag
```

```
Lab_Router(config-if)#ipv6 dhcp server STATEFUL_3
```

```
Lab_Router(config-if)#ipv6 ospf 10 area 0
```

```
Lab_Router(config-if)#no shutdown
```

```
Lab_Router(config-if)#exit
```

```
Lab_Router(config-if)#interface GigabitEthernet 0/0/1
```

```
Lab_Router(config-if)#ipv6 address 2023:2130:2787:5::1/64
```

```
Lab_Router(config-if)#ipv6 address FE80::4:1 link-local
```

```
Lab_Router(config-if)#ipv6 nd managed-config-flag
```

```
Lab_Router(config-if)#ipv6 dhcp server STATEFUL_4
```

```
Lab_Router(config-if)#ipv6 ospf 10 area 0
```

```
Lab_Router(config-if)#no shutdown
```

```
Lab_Router(config-if)#exit
```

```
Lab_Router(config)#interface Serial 0/1/0
```

```
Lab_Router(config-if)#ipv6 address 2023:2130:2787:11::2/64
```

```
Lab_Router(config-if)#ipv6 address FE80::5:4 link-local
```

```
Lab_Router(config-if)#ipv6 enable
```

```
Lab_Router(config-if)#ipv6 ospf 10 area 0
```

```
Lab_Router(config-if)#no shutdown
```

```
Lab_Router(config-if)#end
```

```
Lab_Router#copy run start
```

Internal_Router

```
Router>enable
```

```
Router#configure terminal
```

```
Router(config)#hostname Internal_Router
```

```
Internal_Router(config)#ipv6 unicast-routing
```

```
Internal_Router(config)#ipv6 dhcp pool STATEFUL_5
```

```
Internal_Router(config-dhcpv6)#address prefix  
2023:2130:2787:7::/64
```

```
Internal_Router(config-dhcpv6)#dns-server 2023:2130:2787:1::300
```

```
Internal_Router(config-dhcpv6)#domain-name Classroom.com
```

```
Internal_Router(config-dhcpv6)#exit
```

```
Internal_Router(config)#ipv6 dhcp pool STATEFUL_6
```

```
Internal_Router(config-dhcpv6)#address prefix  
2023:2130:2787:8::/64
```

```
Internal_Router(config-dhcpv6)#dns-server 2023:2130:2787:1::300
```

```
Internal_Router(config-dhcpv6)#domain-name Tutor_admin.com
```

```
Internal_Router(config-dhcpv6)#exit
```

```
Internal_Router(config)#ipv6 router ospf 10
```

```
Internal_Router(config-rtr)#router-id 3.3.3.3
```

```
Internal_Router(config-rtr)#exit
```

```
Internal_Router(config)#interface GigabitEthernet 0/0/0
```

```
Internal_Router(config-if)#ipv6 address 2023:2130:2787:7::1/64
```

```
Internal_Router(config-if)#ipv6 address FE80::8:1 link-local
```

```
Internal_Router(config-if)#ipv6 nd managed-config-flag
```

```
Internal_Router(config-if)#ipv6 dhcp server STATEFUL_5
```

```
Internal_Router(config-if)#ipv6 ospf 10 area 0
```

```
Internal_Router(config-if)#no shutdown
```

```
Internal_Router(config-if)#exit
```

```
Internal_Router(config)#interface GigabitEthernet 0/0/1
```

```
Internal_Router(config-if)#ipv6 address 2023:2130:2787:8::1/64
```

```
Internal_Router(config-if)#ipv6 address FE80::9:1 link-local
```

```
Internal_Router(config-if)#ipv6 nd managed-config-flag
```

```
Internal_Router(config-if)#ipv6 dhcp server STATEFUL_6
```

```
Internal_Router(config-if)#ipv6 ospf 10 area 0
```

```
Internal_Router(config-if)#no shutdown
```

```
Internal_Router(config-if)#exit
```

```
Internal_Router(config)#interface GigabitEthernet 0/0/0
```

```
Internal_Router(config-if)#ipv6 address 2023:2130:2787:7::1/64
```

```
Internal_Router(config-if)#ipv6 address FE80::8:1 link-local
```

```
Internal_Router(config-if)#ipv6 nd managed-config-flag
```

```
Internal_Router(config-if)#ipv6 dhcp server STATEFUL_5
```

```
Internal_Router(config-if)#ipv6 ospf 10 area 0
```

```
Internal_Router(config-if)#no shutdown
```

```
Internal_Router(config-if)#exit
```

```
Internal_Router(config)#interface Serial 0/2/0
```

```
Internal_Router(config-if)#ipv6 address 2023:2130:2787:12::2/64
```

```
Internal_Router(config-if)#ipv6 address FE80::5:6 link-local
```

```
Internal_Router(config-if)#ipv6 enable
```

```
Internal_Router(config-if)#ipv6 ospf 10 area 0
```

```
Internal_Router(config-if)#no shutdown
```

```
Internal_Router(config-if)#end
```

```
Internal_Router#copy run start
```

TM router

```
Router>enable
```

```
Router#configure terminal
```

```
Router(config)#hostname TM_Router
```

```
TM_Router(config)#ipv6 unicast-routing
```

```
TM_Router(config)#ipv6 dhcp pool STATEFUL_7
```

```
TM_Router(config-dhcpv6)#address prefix 2023:2130:2787:9::/64
```

```
TM_Router(config-dhcpv6)#dns-server 2023:2130:2787:1::300
```

```
TM_Router(config-dhcpv6)#domain-name TM_Server.com
```

```
TM_Router(config-dhcpv6)#exit
```

```
TM_Router(config)#ipv6 router ospf 10
```

```
TM_Router(config-rtr)#router-id 4.4.4.4
```

```
TM_Router(config-rtr)#exit
```

```
TM_Router(config)#interface GigabitEthernet 0/0/0
```

```
TM_Router(config-if)#ipv6 address 2023:2130:2787:9::1/64
```

```
TM_Router(config-if)#ipv6 address FE80::10:1 link-local
```

```
TM_Router(config-if)#ipv6 nd managed-config-flag
```

```
TM_Router(config-if)#ipv6 dhcp server STATEFUL_7
```

```
TM_Router(config-if)#ipv6 ospf 10 area 0
```

```
TM_Router(config-if)#no shutdown
```

```
TM_Router(config-if)#exit
```

```
TM_Router(config)#interface Serial 0/1/0
```

```
TM_Router(config-if)#ipv6 address 2023:2130:2787:13::1/64
```

```
TM_Router(config-if)#ipv6 address FE80::5:7 link-local
```

```
TM_Router(config-if)#ipv6 enable
```

```
TM_Router(config-if)#ipv6 ospf 10 area 0
```

```
TM_Router(config-if)#no shutdown
```

```
TM_Router(config-if)#end  
TM_Router#copy run start
```

5. IPV6 ROUTER CONFIGURATION

Main Router

```
Main_Router#show ipv6 route
IPv6 Routing Table - 19 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B -
BGP
U - Per-user Static route, M - MIPv6
I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS
summary
ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr -
Redirect
O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 -
OSPF ext 2
ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
D - EIGRP, EX - EIGRP external
O 2023:1110:319:1::/64 [110/65]
via FE80::5:6, Serial0/2/1
O 2023:1110:319:2::/64 [110/65]
via FE80::5:6, Serial0/2/1
C 2023:2130:2787:1::/64 [0/0]
via GigabitEthernet0/0/0, directly connected
L 2023:2130:2787:1::1/128 [0/0]
via GigabitEthernet0/0/0, receive
C 2023:2130:2787:2::/64 [0/0]
via GigabitEthernet0/0/1, directly connected
L 2023:2130:2787:2::1/128 [0/0]
via GigabitEthernet0/0/1, receive
O 2023:2130:2787:3::/64 [110/65]
via FE80::5:4, Serial0/1/1
O 2023:2130:2787:5::/64 [110/65]
via FE80::5:4, Serial0/1/1
O 2023:2130:2787:7::/64 [110/65]
via FE80::5:6, Serial0/2/1
O 2023:2130:2787:8::/64 [110/65]
via FE80::5:6, Serial0/2/1
O 2023:2130:2787:9::/64 [110/65]
via FE80::5:7, Serial0/1/0
C 2023:2130:2787:10::/64 [0/0]
via Serial0/1/0, directly connected
L 2023:2130:2787:10::1/128 [0/0]
via Serial0/1/0, receive
C 2023:2130:2787:11::/64 [0/0]
via Serial0/1/1, directly connected
L 2023:2130:2787:11::1/128 [0/0]
via Serial0/1/1, receive
C 2023:2130:2787:12::/64 [0/0]
via Serial0/2/1, directly connected
L 2023:2130:2787:12::1/128 [0/0]
via Serial0/2/1, receive
O 2023:2130:2787:13::/64 [110/64]
```

```
via
L FF00::/8 [0/0]
via Null0, receive
```

Lab Router

```
Lab_Router#show ipv6 route
IPv6 Routing Table - 17 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
U - Per-user Static route, M - MIPv6
I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS
summary
ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr -
Redirect
O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF
ext 2
ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
D - EIGRP, EX - EIGRP external
O 2023:1110:319:1::/64 [110/129]
via FE80::5:2, Serial0/1/0
O 2023:1110:319:2::/64 [110/129]
via FE80::5:2, Serial0/1/0
O 2023:2130:2787:1::/64 [110/65]
via FE80::5:2, Serial0/1/0
O 2023:2130:2787:2::/64 [110/65]
via FE80::5:2, Serial0/1/0
C 2023:2130:2787:3::/64 [0/0]
via GigabitEthernet0/0/0, directly connected
L 2023:2130:2787:3::1/128 [0/0]
via GigabitEthernet0/0/0, receive
C 2023:2130:2787:5::/64 [0/0]
via GigabitEthernet0/0/1, directly connected
L 2023:2130:2787:5::1/128 [0/0]
via GigabitEthernet0/0/1, receive
O 2023:2130:2787:7::/64 [110/129]
via FE80::5:2, Serial0/1/0
O 2023:2130:2787:8::/64 [110/129]
via FE80::5:2, Serial0/1/0
O 2023:2130:2787:9::/64 [110/129]
via FE80::5:2, Serial0/1/0
O 2023:2130:2787:10::/64 [110/128]
via FE80::5:2, Serial0/1/0
C 2023:2130:2787:11::/64 [0/0]
via Serial0/1/0, directly connected
L 2023:2130:2787:11::2/128 [0/0]
via Serial0/1/0, receive
O 2023:2130:2787:12::/64 [110/128]
via FE80::5:2, Serial0/1/0
O 2023:2130:2787:13::/64 [110/128]
via FE80::5:2, Serial0/1/0
L FF00::/8 [0/0]
```

via Null0, receive

Internal Router

```
Internal_Router#show ipv6 route
IPv6 Routing Table - 19 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
U - Per-user Static route, M - MIPv6
I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS
summary
ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr -
Redirect
O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF
ext 2
ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
D - EIGRP, EX - EIGRP external
C 2023:1110:319:1::/64 [0/0]
via GigabitEthernet0/0/0, directly connected
L 2023:1110:319:1::1/128 [0/0]
via GigabitEthernet0/0/0, receive
C 2023:1110:319:2::/64 [0/0]
via GigabitEthernet0/0/1, directly connected
L 2023:1110:319:2::1/128 [0/0]
via GigabitEthernet0/0/1, receive
O 2023:2130:2787:1::/64 [110/65]
via FE80::5:3, Serial0/2/0
O 2023:2130:2787:2::/64 [110/65]
via FE80::5:3, Serial0/2/0
O 2023:2130:2787:3::/64 [110/129]
via FE80::5:3, Serial0/2/0
O 2023:2130:2787:5::/64 [110/129]
via FE80::5:3, Serial0/2/0
C 2023:2130:2787:7::/64 [0/0]
via GigabitEthernet0/0/0, directly connected
L 2023:2130:2787:7::1/128 [0/0]
via GigabitEthernet0/0/0, receive
C 2023:2130:2787:8::/64 [0/0]
via GigabitEthernet0/0/1, directly connected
L 2023:2130:2787:8::1/128 [0/0]
via GigabitEthernet0/0/1, receive
O 2023:2130:2787:9::/64 [110/129]
via FE80::5:3, Serial0/2/0
O 2023:2130:2787:10::/64 [110/128]
via FE80::5:3, Serial0/2/0
O 2023:2130:2787:11::/64 [110/128]
via FE80::5:3, Serial0/2/0
C 2023:2130:2787:12::/64 [0/0]
via Serial0/2/0, directly connected
L 2023:2130:2787:12::2/128 [0/0]
via Serial0/2/0, receive
O 2023:2130:2787:13::/64 [110/128]
```

```
via FE80::5:3, Serial0/2/0
L FF00::/8 [0/0]
via Null0, receive
```

TM Router

```
TM_Router#show ipv6 route
IPv6 Routing Table - 16 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
U - Per-user Static route, M - MIPv6
I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS
summary
ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr -
Redirect
O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF
ext 2
ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
D - EIGRP, EX - EIGRP external
O 2023:1110:319:1::/64 [110/129]
via FE80::5:1, Serial0/1/0
O 2023:1110:319:2::/64 [110/129]
via FE80::5:1, Serial0/1/0
O 2023:2130:2787:1::/64 [110/65]
via FE80::5:1, Serial0/1/0
O 2023:2130:2787:2::/64 [110/65]
via FE80::5:1, Serial0/1/0
O 2023:2130:2787:3::/64 [110/129]
via FE80::5:1, Serial0/1/0
O 2023:2130:2787:5::/64 [110/129]
via FE80::5:1, Serial0/1/0
O 2023:2130:2787:7::/64 [110/129]
via FE80::5:1, Serial0/1/0
O 2023:2130:2787:8::/64 [110/129]
via FE80::5:1, Serial0/1/0
C 2023:2130:2787:9::/64 [0/0]
via GigabitEthernet0/0/0, directly connected
L 2023:2130:2787:9::1/128 [0/0]
via GigabitEthernet0/0/0, receive
O 2023:2130:2787:10::/64 [110/64]
via
O 2023:2130:2787:11::/64 [110/128]
via FE80::5:1, Serial0/1/0
O 2023:2130:2787:12::/64 [110/128]
via FE80::5:1, Serial0/1/0
C 2023:2130:2787:13::/64 [0/0]
via Serial0/1/0, directly connected
L 2023:2130:2787:13::1/128 [0/0]
via Serial0/1/0, receive
L FF00::/8 [0/0]
via Null0, receive
```

6. DHCP SERVER CONFIGURATION

Main_Router

```
Main_Router>enable
```

```
Main_Router#configure terminal
```

```
Main_Router(config)#ipv6 unicast-routing
```

```
Main_Router(config)#ipv6 dhcp pool STATEFUL_1
```

```
Main_Router(config-dhcpv6)#address prefix 2023:2130:2787:1::/64
```

```
Main_Router(config-dhcpv6)#dns-server 2023:2130:2787:1::300
```

```
Main_Router(config-dhcpv6)#domain-name Main_server.com
```

```
Main_Router(config-dhcpv6)#exit
```

```
Main_Router(config)#ipv6 dhcp pool STATEFUL_2
```

```
Main_Router(config-dhcpv6)#address prefix 2023:2130:2787:2::/64
```

```
Main_Router(config-dhcpv6)#dns-server 2023:2130:2787:1::300
```

```
Main_Router(config-dhcpv6)#domain-name Main_wifi.com
```

```
Main_Router(config-dhcpv6)#exit
```

```
Main_Router(config)#interface GigabitEthernet 0/0/0
```

```
Main_Router(config-if)#ipv6 nd managed-config-flag
```

```
Main_Router(config-if)#ipv6 dhcp server STATEFUL_1
```

```
Main_Router(config-if)#no shutdown
```

```
Main_Router(config-if)#exit
```

```
Main_Router(config)#interface GigabitEthernet 0/0/1
```

```
Main_Router(config-if)#ipv6 nd managed-config-flag
```

```
Main_Router(config-if)#ipv6 dhcp server STATEFUL_2
```

```
Main_Router(config-if)#no shutdown
```

```
Main_Router(config-if)#exit
```

Lab_Router

```
Lab_Router>enable
```

```
Lab_Router#configure terminal
```

```
Lab_Router(config)#ipv6 unicast-routing
```

```
Lab_Router(config)#ipv6 dhcp pool STATEFUL_3
```

```
Lab_Router(config-dhcpv6)#address prefix 2023:2130:2787:3::/64
```

```
Lab_Router(config-dhcpv6)#dns-server 2023:2130:2787:1::300
```

```
Lab_Router(config-dhcpv6)#domain-name Lab_1.com
```

```
Lab_Router(config-dhcpv6)#exit
```

```
Lab_Router(config)#ipv6 dhcp pool STATEFUL_4
```

```
Lab_Router(config-dhcpv6)#address prefix 2023:2130:2787:5::/64
```

```
Lab_Router(config-dhcpv6)#dns-server 2023:2130:2787:1::300
```

```
Lab_Router(config-dhcpv6)#domain-name Lab_2.com
```

```
Lab_Router(config-dhcpv6)#exit
```

```
Lab_Router(config)#interface GigabitEthernet 0/0/0
```

```
Lab_Router(config-if)#ipv6 nd managed-config-flag
```

```
Lab_Router(config-if)#ipv6 dhcp server STATEFUL_3
```

```
Lab_Router(config-if)#no shutdown
```

```
Lab_Router(config-if)#exit
```

```
Lab_Router(config-if)#interface GigabitEthernet 0/0/1
```

```
Lab_Router(config-if)#ipv6 nd managed-config-flag
```

```
Lab_Router(config-if)#ipv6 dhcp server STATEFUL_4
```

```
Lab_Router(config-if)#no shutdown
```

```
Lab_Router(config-if)#exit
```

Internal_Router

```
Internal_Router>enable
Internal_Router#configure terminal
Internal_Router(config)#ipv6 unicast-routing
Internal_Router(config)#ipv6 dhcp pool STATEFUL_5
Internal_Router(config-dhcpv6)#address prefix
2023:2130:2787:7::/64
Internal_Router(config-dhcpv6)#dns-server 2023:2130:2787:1::300
Internal_Router(config-dhcpv6)#domain-name Classroom.com
Internal_Router(config-dhcpv6)#exit

Internal_Router(config)#ipv6 dhcp pool STATEFUL_6
Internal_Router(config-dhcpv6)#address prefix
2023:2130:2787:8::/64
Internal_Router(config-dhcpv6)#dns-server 2023:2130:2787:1::300
Internal_Router(config-dhcpv6)#domain-name Tutor_admin.com
Internal_Router(config-dhcpv6)#exit

Internal_Router(config)#interface GigabitEthernet 0/0/0
Internal_Router(config-if)#ipv6 nd managed-config-flag
Internal_Router(config-if)#ipv6 dhcp server STATEFUL_5
Internal_Router(config-if)#no shutdown
Internal_Router(config-if)#exit

Internal_Router(config)#interface GigabitEthernet 0/0/1
Internal_Router(config-if)#ipv6 nd managed-config-flag
Internal_Router(config-if)#ipv6 dhcp server STATEFUL_6
Internal_Router(config-if)#no shutdown
Internal_Router(config-if)#exit
```

TM Router

```
TM_Router>enable
```

```
TM_Router#configure terminal
```

```
TM_Router(config)#ipv6 unicast-routing
```

```
TM_Router(config)#ipv6 dhcp pool STATEFUL_7
```

```
TM_Router(config-dhcpv6)#address prefix 2023:2130:2787:9::/64
```

```
TM_Router(config-dhcpv6)#dns-server 2023:2130:2787:1::300
```

```
TM_Router(config-dhcpv6)#domain-name TM_Server.com
```

```
TM_Router(config-dhcpv6)#exit
```

```
TM_Router(config)#interface GigabitEthernet 0/0/0
```

```
TM_Router(config-if)#ipv6 nd managed-config-flag
```

```
TM_Router(config-if)#ipv6 dhcp server STATEFUL_7
```

```
TM_Router(config-if)#no shutdown
```

```
TM_Router(config-if)#exit
```

7. OSPF CONFIGURATION

Main Router

```
Main_Router>enable
Main_Router#configure terminal
Main_Router(config)#ipv6 router ospf 10
Main_Router(config-rtr)#router-id 1.1.1.1
Main_Router(config-rtr)#exit

Main_Router(config)#interface GigabitEthernet 0/0/0
Main_Router(config-if)#ipv6 ospf 10 area 0
Main_Router(config-if)#no shutdown
Main_Router(config-if)#exit

Main_Router(config)#interface GigabitEthernet 0/0/1
Main_Router(config-if)#ipv6 ospf 10 area 0
Main_Router(config-if)#no shutdown
Main_Router(config-if)#exit

Main_Router(config)#interface Serial 0/1/0
Main_Router(config-if)#ipv6 ospf 10 area 0
Main_Router(config-if)#no shutdown
Main_Router(config-if)#exit

Main_Router(config)#interface serial 0/1/1
Main_Router(config-if)#ipv6 ospf 10 area 0
Main_Router(config-if)#no shutdown
Main_Router(config-if)#exit

Main_Router(config)#interface Serial 0/2/1
Main_Router(config-if)#ipv6 ospf 10 area 0
Main_Router(config-if)#no shutdown
Main_Router(config-if)#exit
```

Lab Router

```
Lab_Router>enable
Lab_Router#configure terminal
Lab_Router(config)#ipv6 router ospf 10
Lab_Router(config-rtr)#router-id 2.2.2.2
Lab_Router(config-rtr)#exit

Lab_Router(config)#interface GigabitEthernet 0/0/0
Lab_Router(config-if)#ipv6 ospf 10 area 0
Lab_Router(config-if)#no shutdown
Lab_Router(config-if)#exit

Lab_Router(config-if)#interface GigabitEthernet 0/0/1
Lab_Router(config-if)#ipv6 ospf 10 area 0
Lab_Router(config-if)#no shutdown
```

```
Lab_Router(config-if)#exit
```

```
Lab_Router(config)#interface Serial 0/1/0
Lab_Router(config-if)#ipv6 ospf 10 area 0
Lab_Router(config-if)#no shutdown
Lab_Router(config-if)#end
```

Internal Router

```
Internal_Router>enable
Internal_Router#configure terminal
Internal_Router(config)#ipv6 router ospf 10
Internal_Router(config-rtr)#router-id 3.3.3.3
Internal_Router(config-rtr)#exit
```

```
Internal_Router(config)#interface GigabitEthernet 0/0/0
Internal_Router(config-if)#ipv6 ospf 10 area 0
Internal_Router(config-if)#no shutdown
Internal_Router(config-if)#exit
```

```
Internal_Router(config)#interface GigabitEthernet 0/0/1
Internal_Router(config-if)#ipv6 ospf 10 area 0
Internal_Router(config-if)#no shutdown
Internal_Router(config-if)#exit
```

```
Internal_Router(config)#interface GigabitEthernet 0/0/0
Internal_Router(config-if)#ipv6 ospf 10 area 0
Internal_Router(config-if)#no shutdown
Internal_Router(config-if)#exit
```

```
Internal_Router(config)#interface Serial 0/2/0
Internal_Router(config-if)#ipv6 ospf 10 area 0
Internal_Router(config-if)#no shutdown
Internal_Router(config-if)#end
```

TM Router

```
TM_Router>enable
TM_Router#configure terminal
TM_Router(config)#ipv6 router ospf 10
TM_Router(config-rtr)#router-id 4.4.4.4
TM_Router(config-rtr)#exit
```

```
TM_Router(config)#interface GigabitEthernet 0/0/0
TM_Router(config-if)#ipv6 ospf 10 area 0
TM_Router(config-if)#no shutdown
TM_Router(config-if)#exit
```

```
TM_Router(config)#interface Serial 0/1/0
```

```
TM_Router(config-if)#ipv6 ospf 10 area 0
TM_Router(config-if)#no shutdown
TM_Router(config-if)#end
```

8. CLOUD AND ISP ROUTER CONFIGURATION

Main Router

```
Main_Router>enable
Main_Router#configure terminal
Main_Router(config)#int se0/1/0
Main_Router(config-if)#encap frame-relay
Main_Router(config-if)#frame-relay map ipv6
2023:2130:2787:13::1 100
Main_Router(config-if)#frame-relay map ipv6 FE80::5:7 100
broadcast
Main_Router(config-if)#ipv6 add FE80::5:1 link-local
Main_Router(config-if)#ipv6 ospf neighbor FE80::5:7
Main_Router(config-if)#no shut
Main_Router(config-if)#end
Main_Router#copy run start
```

TM Router

```
TM_Router>enable
TM_Router#configure terminal
TM_Router(config)#int se0/1/0
TM_Router(config-if)#encap frame-relay
TM_Router(config-if)#frame-relay map ipv6 2023:2130:2787:10::1
200
TM_Router(config-if)#frame-relay map ipv6 FE80::5:1 200
broadcast
TM_Router(config-if)#ipv6 add FE80::5:7 link-local
TM_Router(config-if)#ipv6 ospf neighbor FE80::5:1
TM_Router(config-if)#no shut
TM_Router(config-if)#end
TM_Router#copy run start
```

9. ACCESS CONTROL LIST

9.1.DENY INTERNET ACCESS TO ADMIN AND TUTOR ROOM

DENY INTERNET ACCESS FOR TUTOR AND ADMIN ROOM

```
Internal_Router>enable
Internal_Router#configure terminal
Internal_Router(config)#IPv6 access-list Tutor_admin
Internal_Router(config-ipv6-acl)#deny tcp
2023:2130:2787:8::1/64 2023:2130:2787:9::1/64 eq www
Internal_Router(config-ipv6-acl)#deny tcp
2023:2130:2787:8::1/64 2023:2130:2787:9::1/64 eq 443
Internal_Router(config-ipv6-acl)#permit icmp any any
Internal_Router(config-ipv6-acl)#permit ipv6 any any
Internal_Router(config-ipv6-acl)#permit udp any any
Internal_Router(config-ipv6-acl)#exit
Internal_Router(config)#interface GigabitEthernet 0/0/1
Internal_Router(config-if)#ipv6 traffic-filter Tutor_admin in
Internal_Router(config-if)#end
Internal_Router#copy run start
```

9.2.DENY INTERNET ACCESS TO WIRELESS NETWORK

DENY INTERNET ACCESS FOR WIRELESS DEVICE

```
Main_Router>enable
Main_Router#configure terminal
Main_Router(config)#IPv6 access-list Wireless
Main_Router(config-ipv6-acl)#deny tcp 2023:2130:2787:2::1/64
2023:2130:2787:9::1/64 eq www
Main_Router(config-ipv6-acl)#deny tcp 2023:2130:2787:2::1/64
2023:2130:2787:9::1/64 eq 443
Main_Router(config-ipv6-acl)#permit icmp any any
Main_Router(config-ipv6-acl)#permit ipv6 any any
Main_Router(config-ipv6-acl)#permit udp any any
Main_Router(config-ipv6-acl)#exit
Main_Router(config)#interface GigabitEthernet 0/0/1
Main_Router(config-if)#ipv6 traffic-filter Wireless in
Main_Router(config-if)#end
Main_Router#copy run start
```

10. WEB SERVER CONFIGURATION

The screenshot shows a web browser window titled "WebServer(mmu.edu.my)". The "Desktop" tab is selected in the top navigation bar. The "IP Configuration" section is active, displaying settings for IPv4, IPv6, and 802.1X.

IP Configuration

☐ DHCP ☒ Static

IPv4 Address:

Subnet Mask:

Default Gateway:

DNS Server:

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: / 64

Link Local Address:

Default Gateway:

DNS Server:

802.1X

☐ Use 802.1X Security

Authentication:

Username:

Password:

Figure 10.1. IP Configuratin for web server

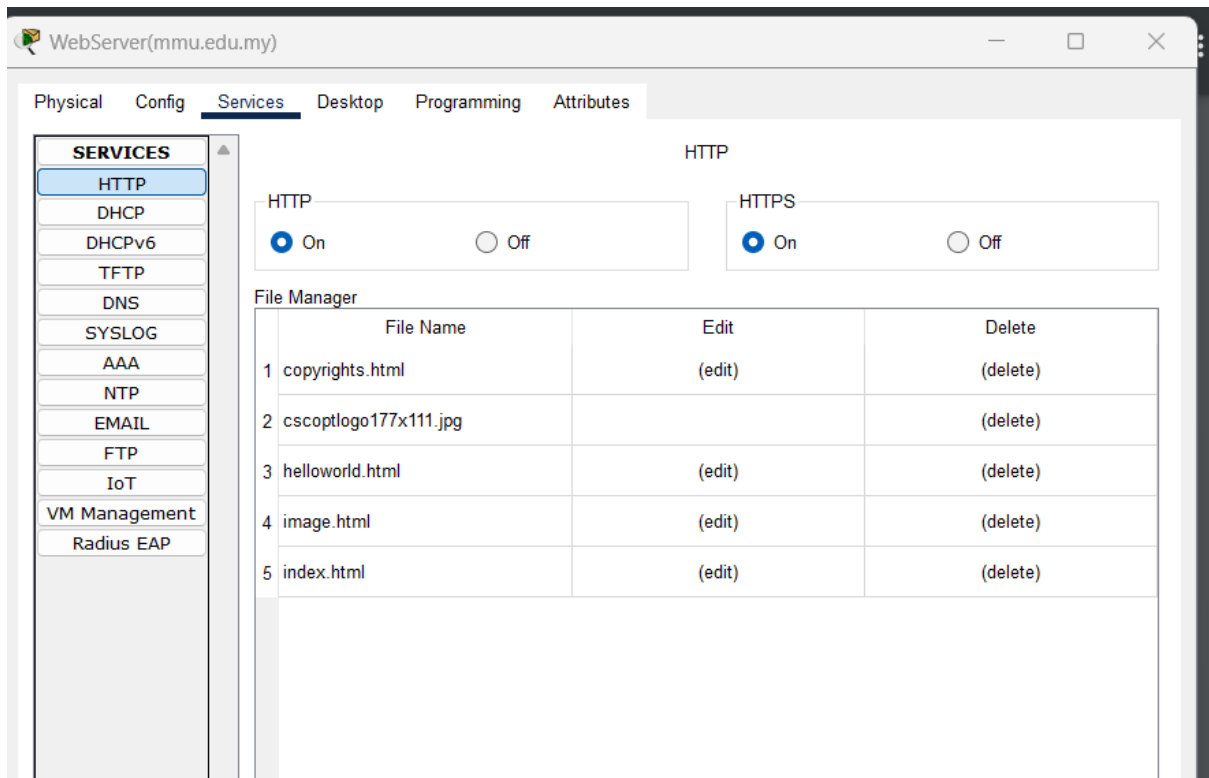


Figure 10.2. HTTP configuration for web server

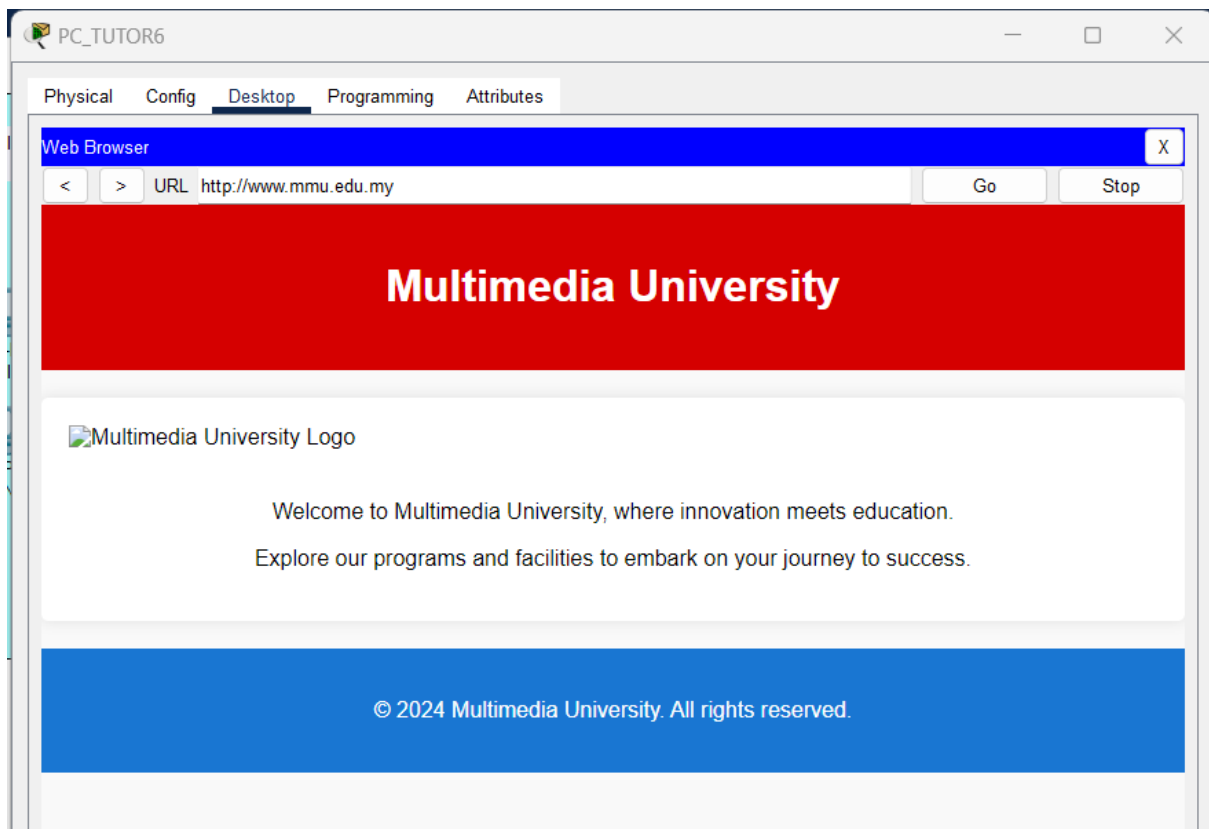


Figure 10.2 Accessing web server from PC6 tutor room

11. FTP Server Configuration

The screenshot shows a web-based configuration interface for an FTP server. The window title is "FTP Server:FTP.com". The navigation tabs are "Physical", "Config", "Services", "Desktop", "Programming", and "Attributes". The "Desktop" tab is selected, and the "IP Configuration" sub-tab is active. The "IP Configuration" section has a blue header bar with a close button (X). It contains two main sections: "IP Configuration" and "IPv6 Configuration".

IP Configuration

- ☐ DHCP
- ☒ Static
- IPv4 Address:
- Subnet Mask:
- Default Gateway:
- DNS Server:

IPv6 Configuration

- ☐ Automatic
- ☒ Static
- IPv6 Address: /
- Link Local Address:
- Default Gateway:
- DNS Server:

802.1X

- ☐ Use 802.1X Security
- Authentication:
- Username:
- Password:

Figure 11.1: IP Configuration of FTP Server

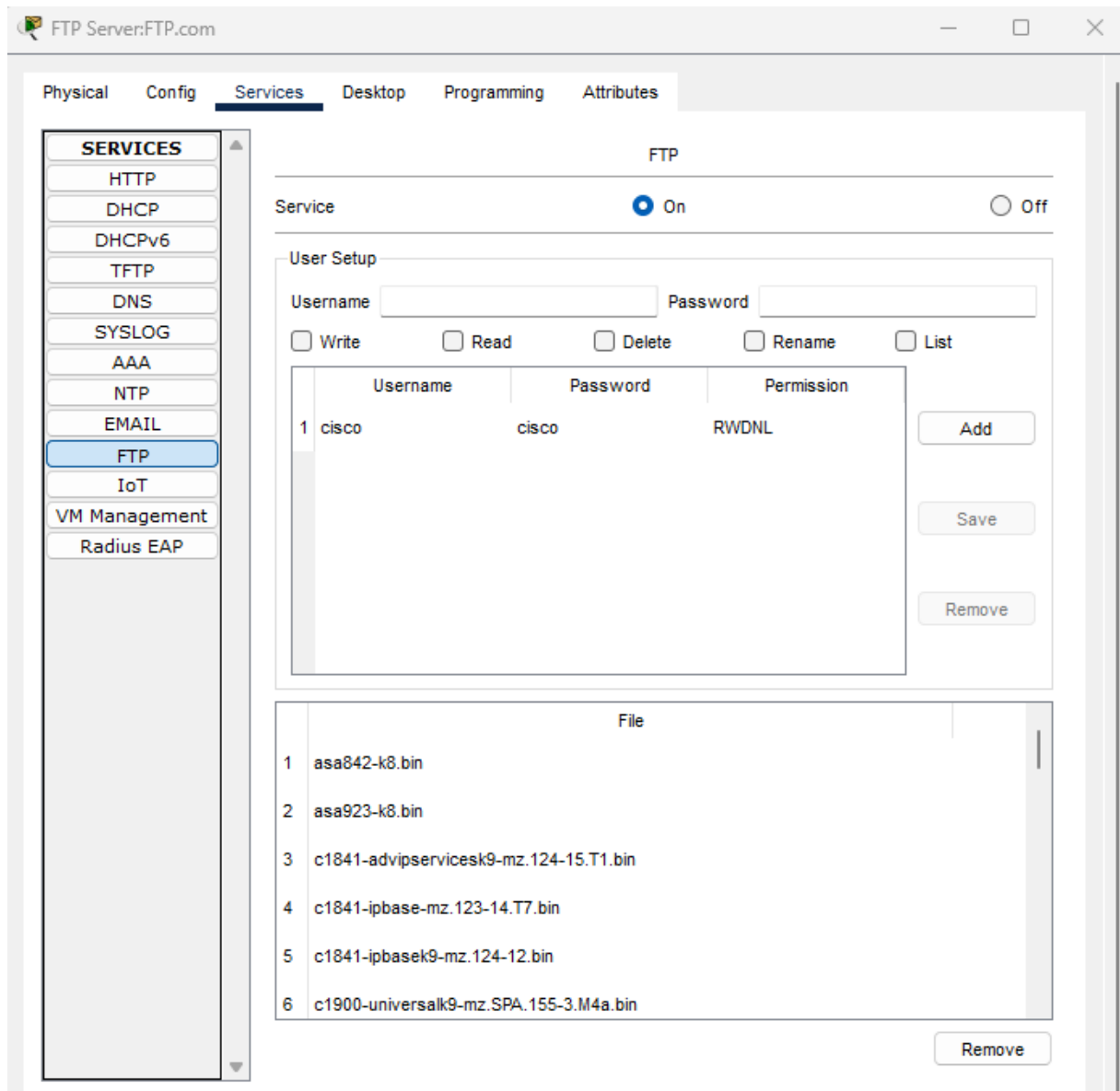


Figure 11.2. FTP Service Configuration

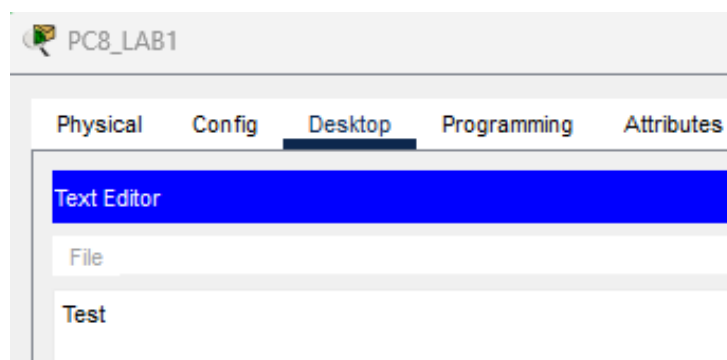


Figure 11.3 : Creating a test file using PC8_Lab1

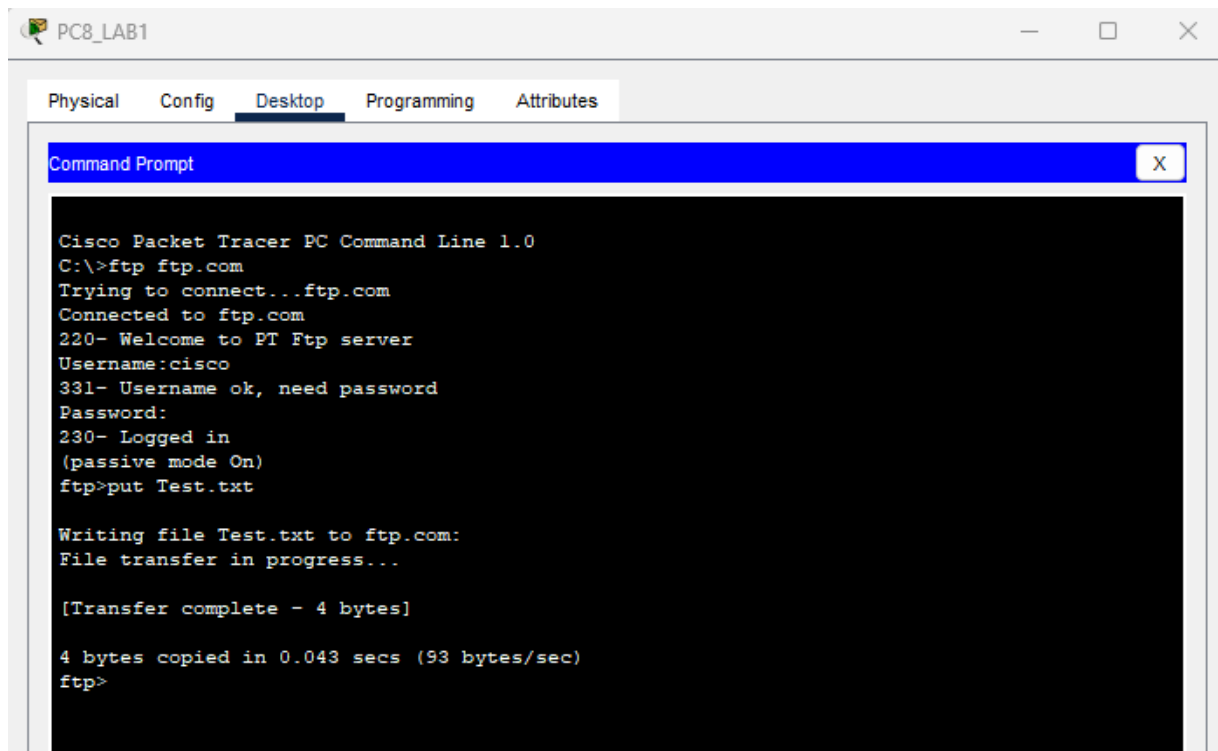


Figure 11.4 : Uploading test file named Test.txt to the File Server from PC8_LAB1

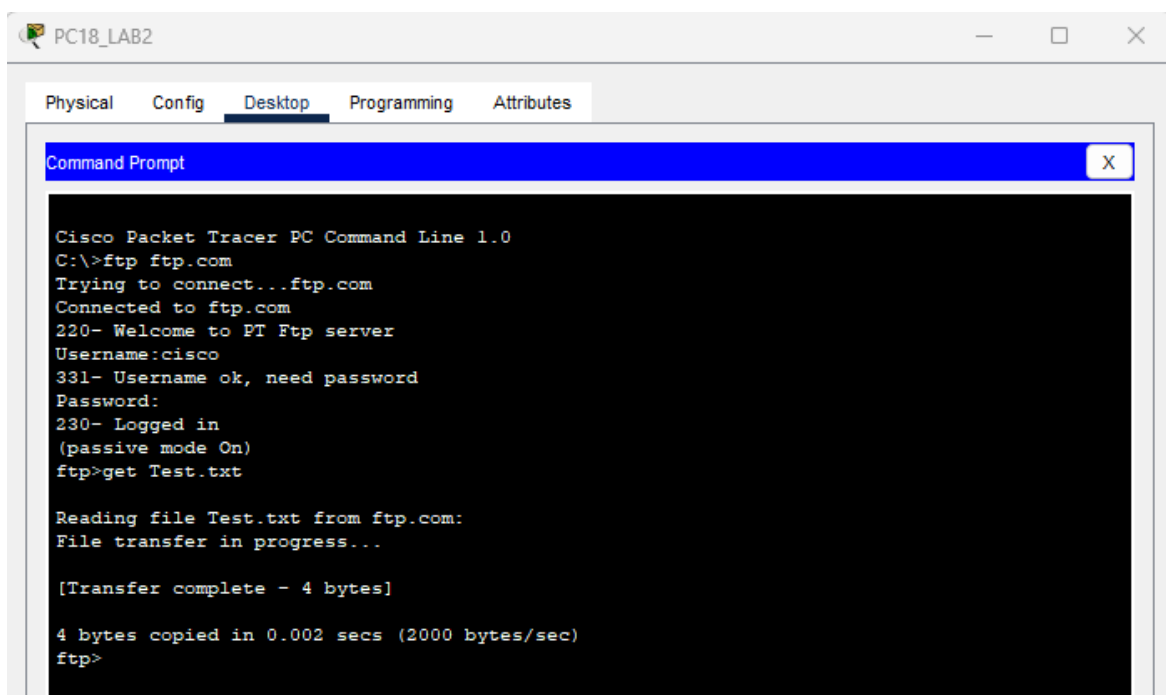


Figure 11.5 : Downloading Test.txt file to PC18_lab2

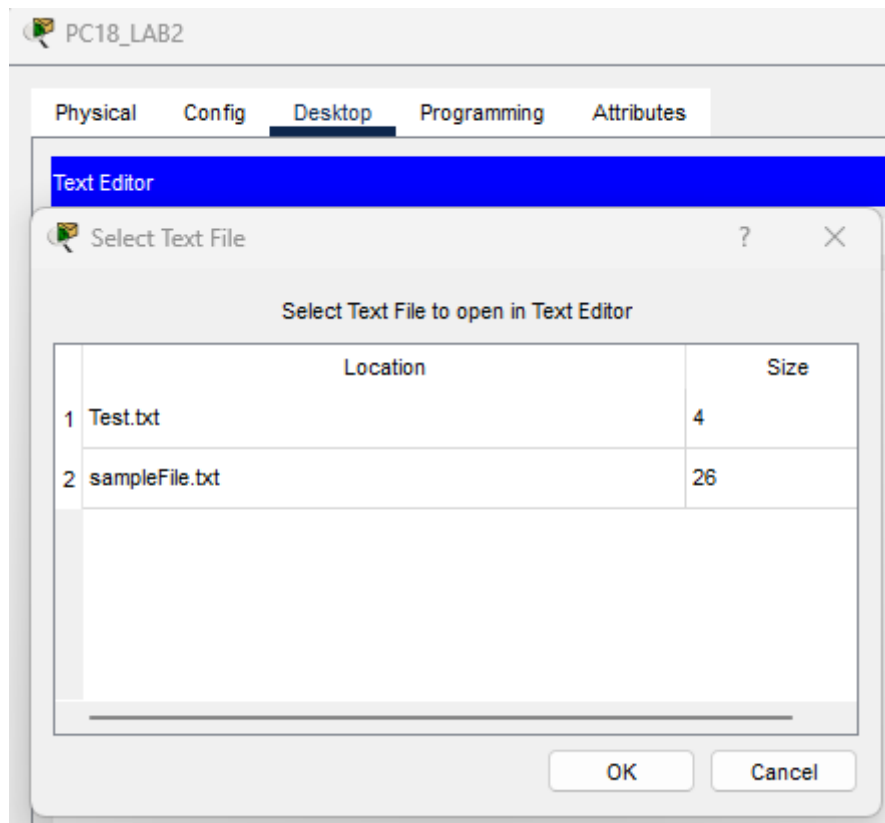


Figure 11.6 : Test.txt file is downloaded to PC18_LAB2

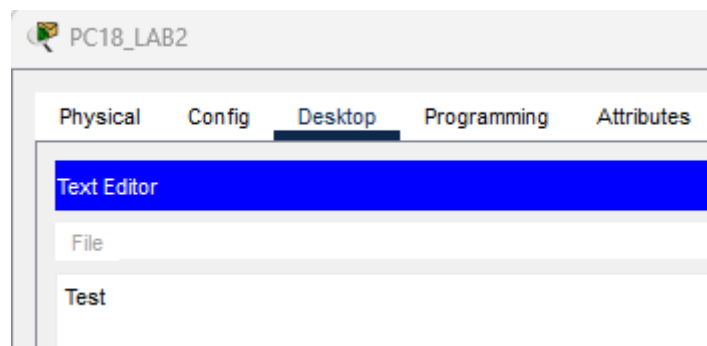


Figure 11.7 : Contents of Test.txt file can be opened and read using PC18_LAB2

12. DNS SERVER CONFIGURATION

The screenshot shows a web-based configuration interface for a DNS server. The window has a title bar 'DNS_Server' and a menu bar with 'Physical', 'Config', 'Services', 'Desktop' (selected), 'Programming', and 'Attributes'. Below the menu bar is a tabbed interface with 'IP Configuration' selected. The 'IP Configuration' section contains two sub-sections: 'IP Configuration' and 'IPv6 Configuration'. The 'IP Configuration' section has radio buttons for 'DHCP' and 'Static' (selected). Below these are input fields for 'IPv4 Address', 'Subnet Mask', 'Default Gateway' (0.0.0.0), and 'DNS Server' (0.0.0.0). The 'IPv6 Configuration' section has radio buttons for 'Automatic' and 'Static' (selected). Below these are input fields for 'IPv6 Address' (2023:2130:2787:1::300 / 64), 'Link Local Address' (FE80::2D0:D3FF:FEDC:5EAA), 'Default Gateway' (2023:2130:2787:1::1), and 'DNS Server' (2023:2130:2787:1::300). Below the IPv6 section is the '802.1X' section, which includes a checkbox for 'Use 802.1X Security' (unchecked), a dropdown for 'Authentication' (MD5), and input fields for 'Username' and 'Password'.

Physical Config Services **Desktop** Programming Attributes

IP Configuration X

IP Configuration

☐ DHCP ☒ Static

IPv4 Address

Subnet Mask

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address 2023:2130:2787:1::300 / 64

Link Local Address FE80::2D0:D3FF:FEDC:5EAA

Default Gateway 2023:2130:2787:1::1

DNS Server 2023:2130:2787:1::300

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

Figure 12.1. IP configuration for DNS server

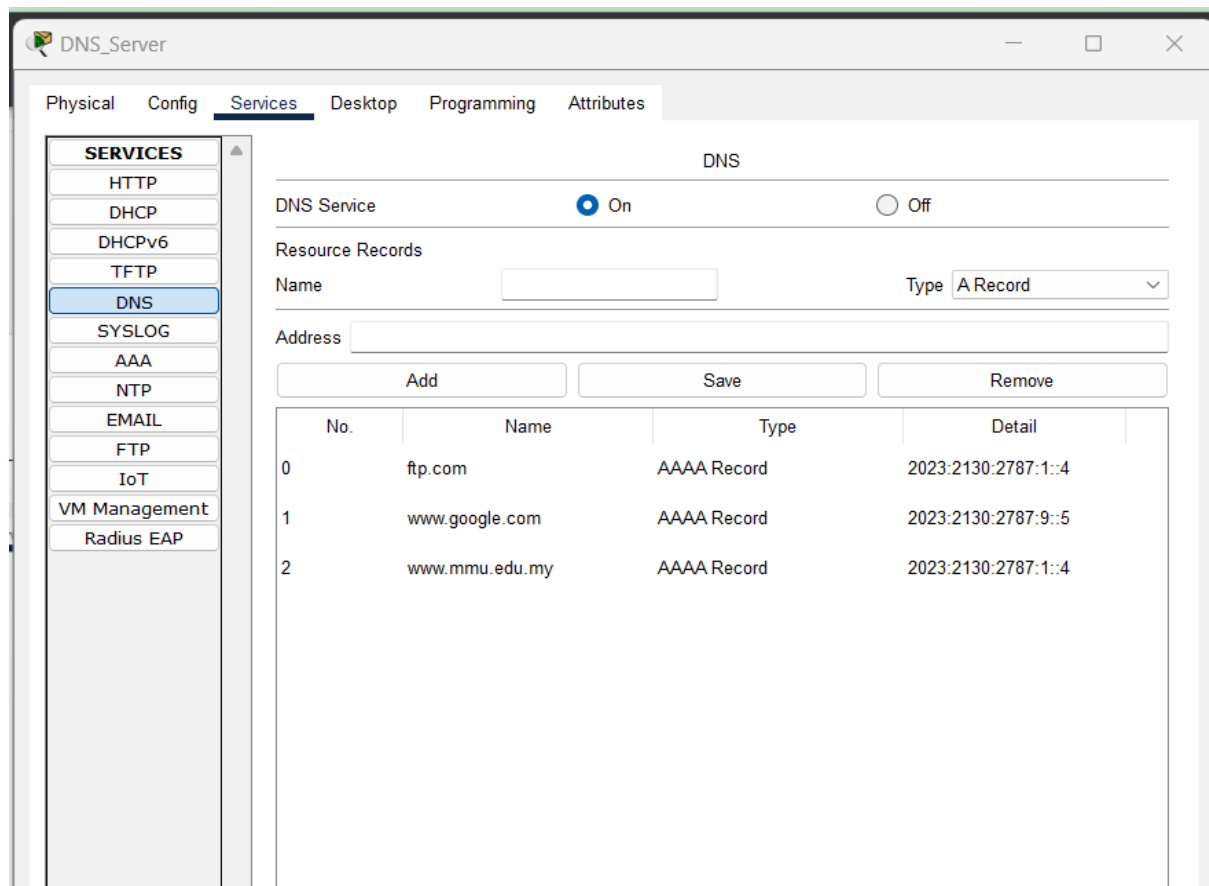


Figure 12.2. DNS services configuraion for DNS Server

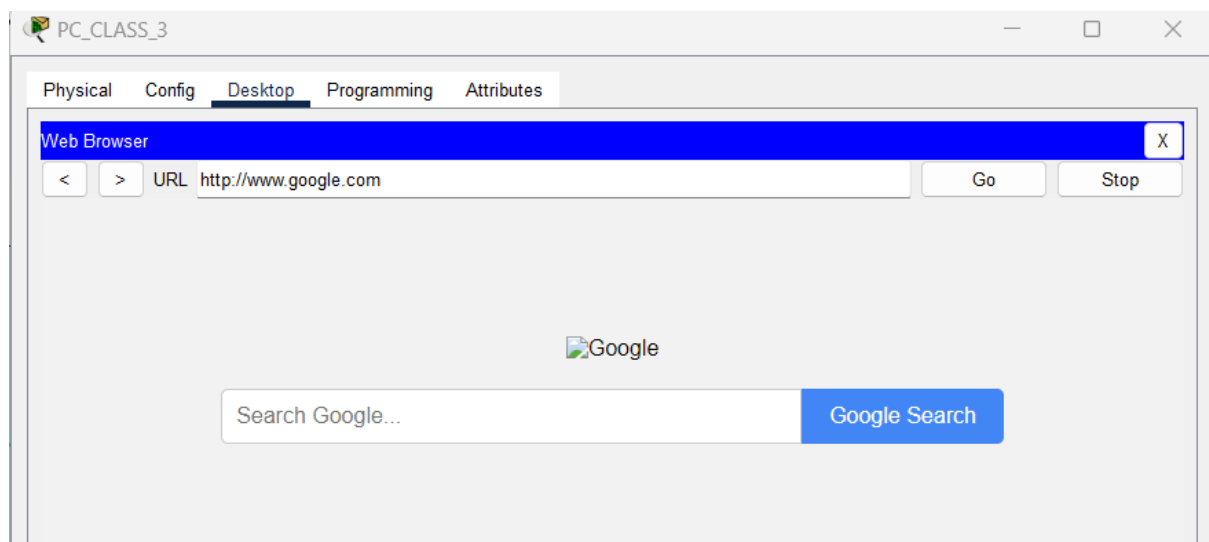


Figure 12.3 Accessing DNS server from PC3 classroom

13. TM SERVER CONFIGURATION

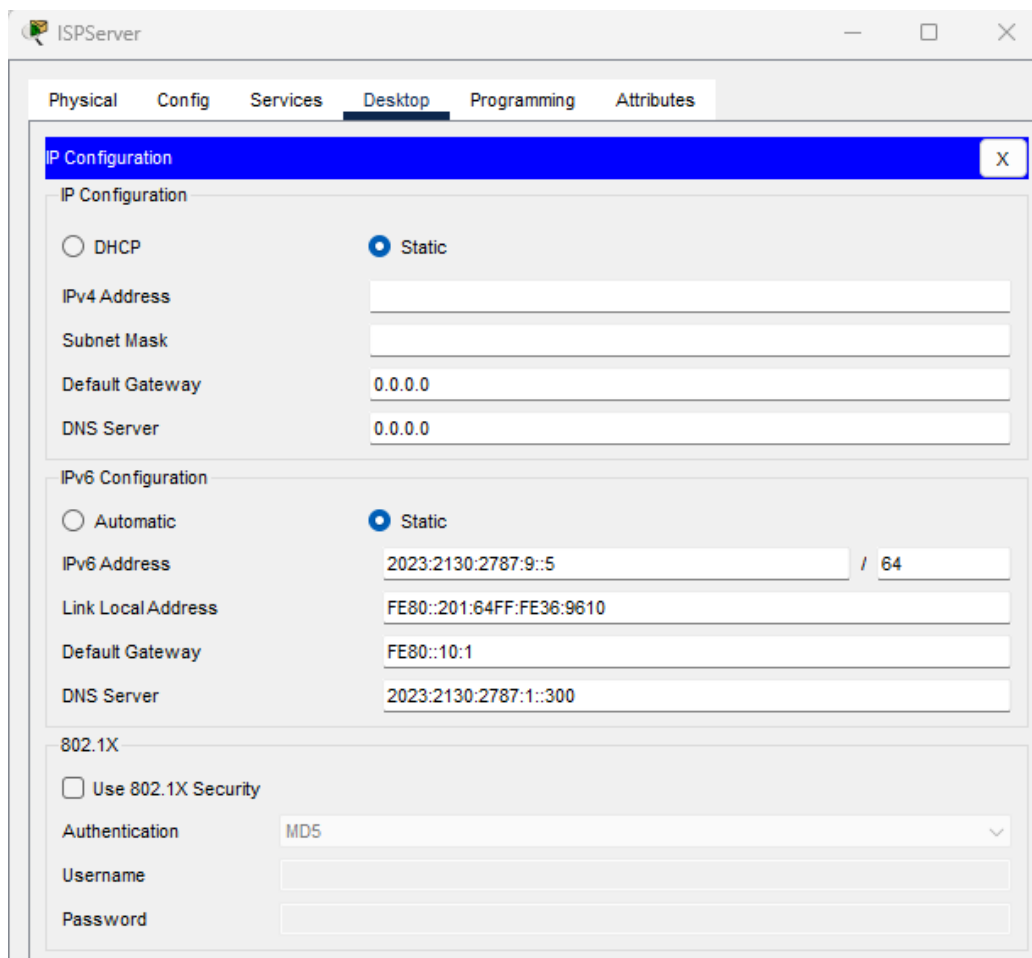


Figure 13.1 : IP Configuration of TM Server(ISP Server)

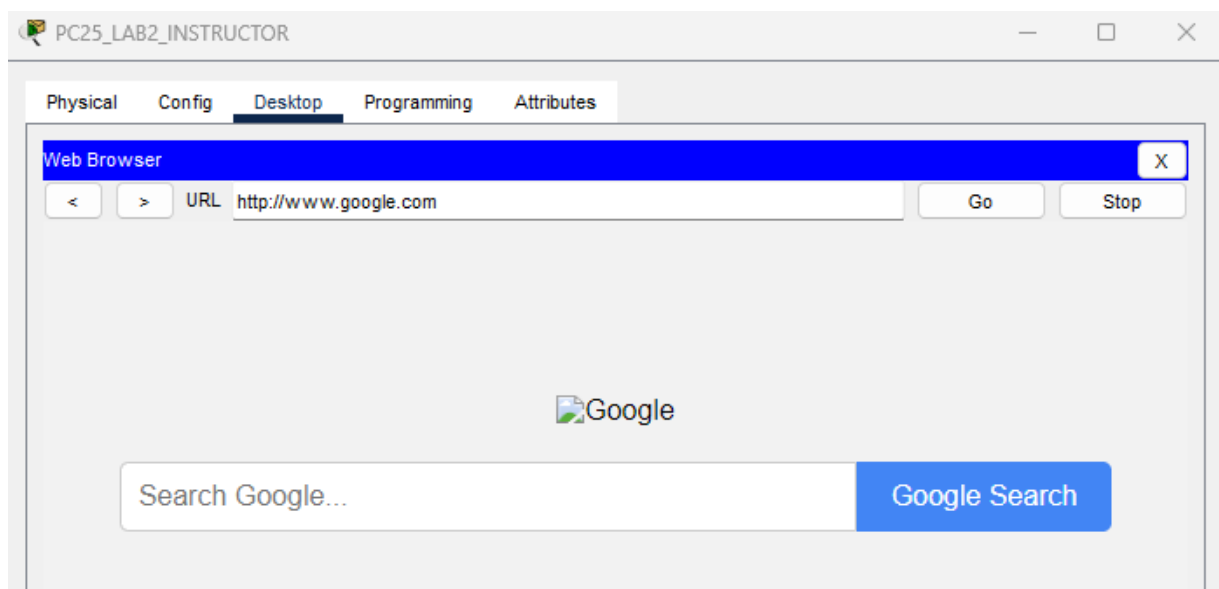


Figure 13.2 : PC25_LAB2_INSTRUCTOR can access www.google.com

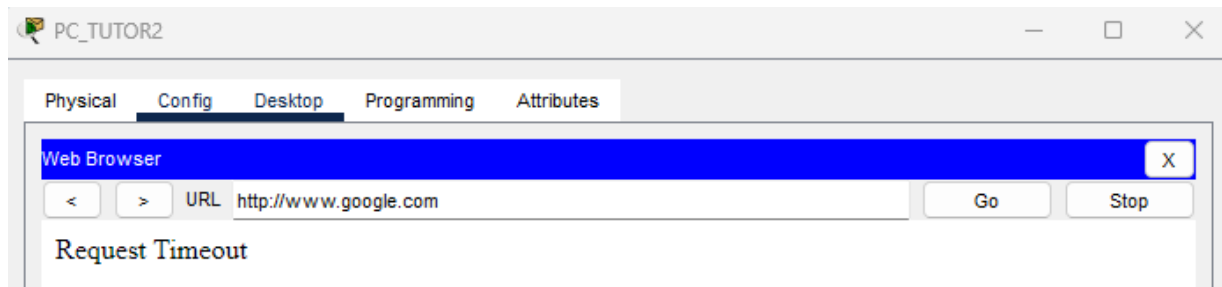


Figure 13.3 : PC_TUTOR2 not allowed access to www.google.com

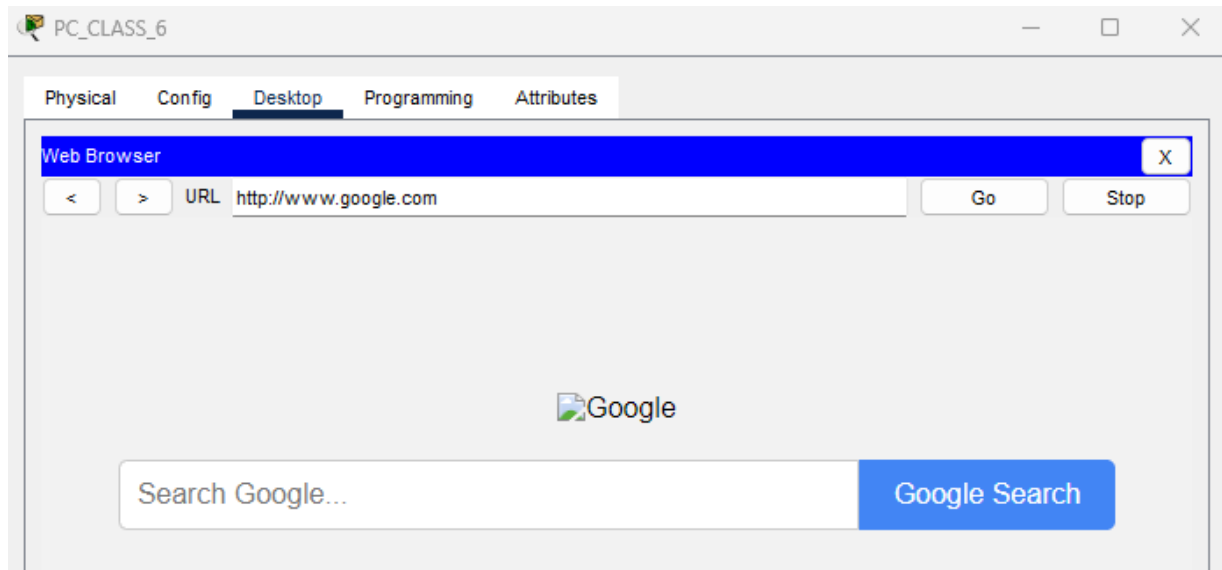


Figure 13.4 : PC_CLASS_6 allowed access to www.google.com

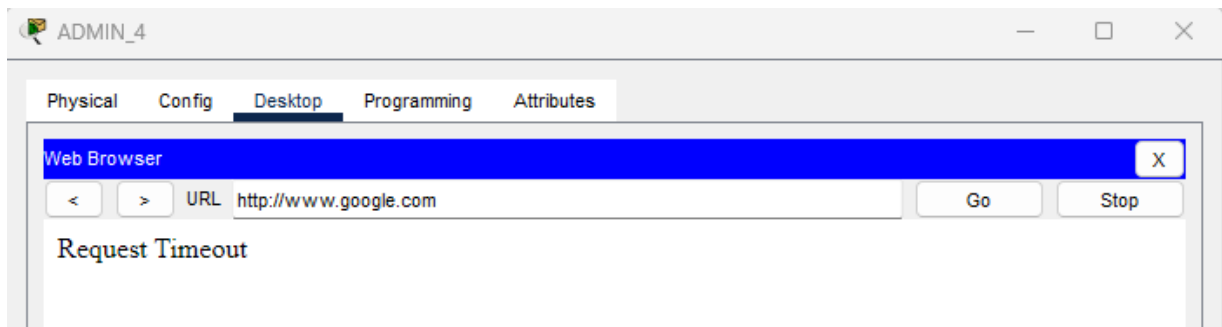


Figure 13.5 : ADMIN_4 not allowed access to www.google.com

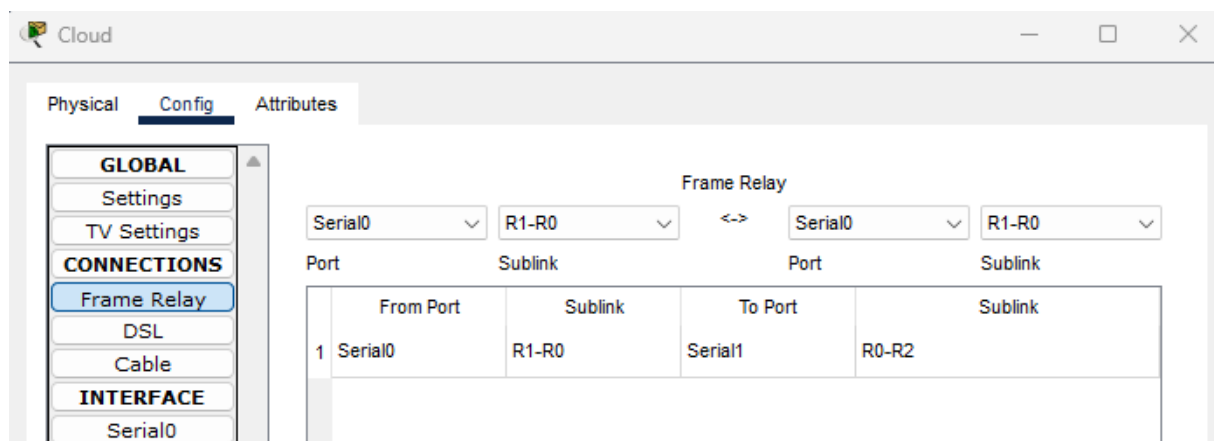


Figure 13.6 : Shows the link between Serial0 and Serial1 in Frame Relay

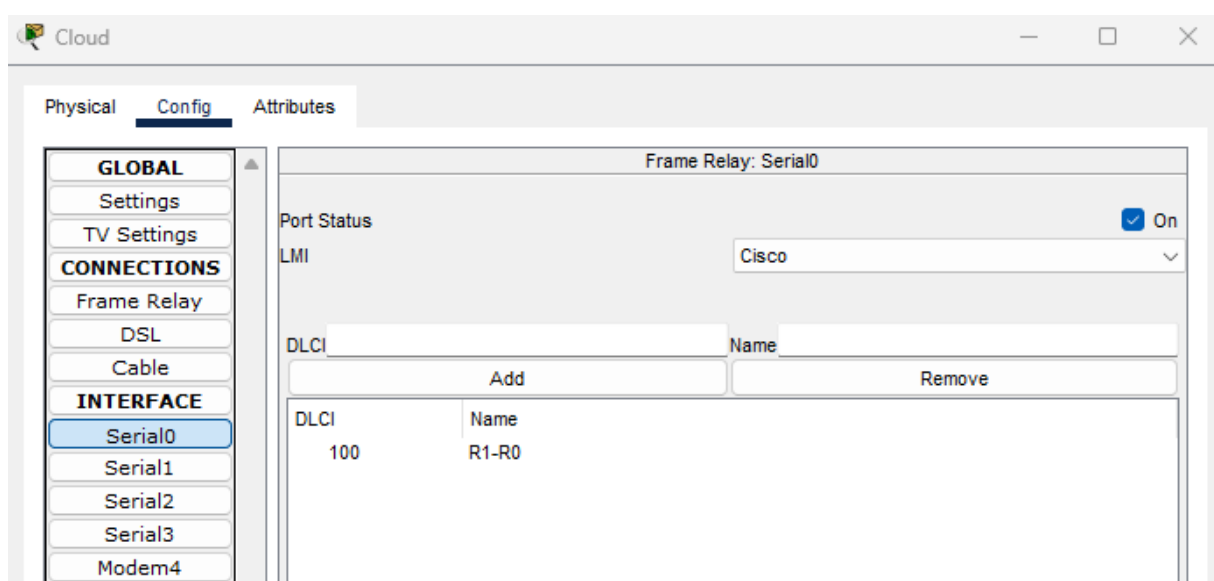


Figure 13.7 : Shows interface for Serial1

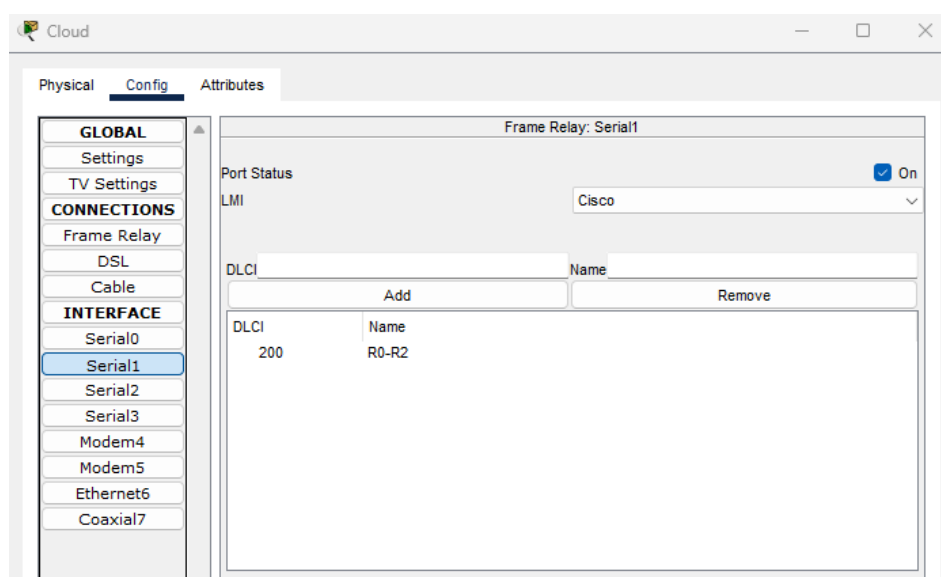
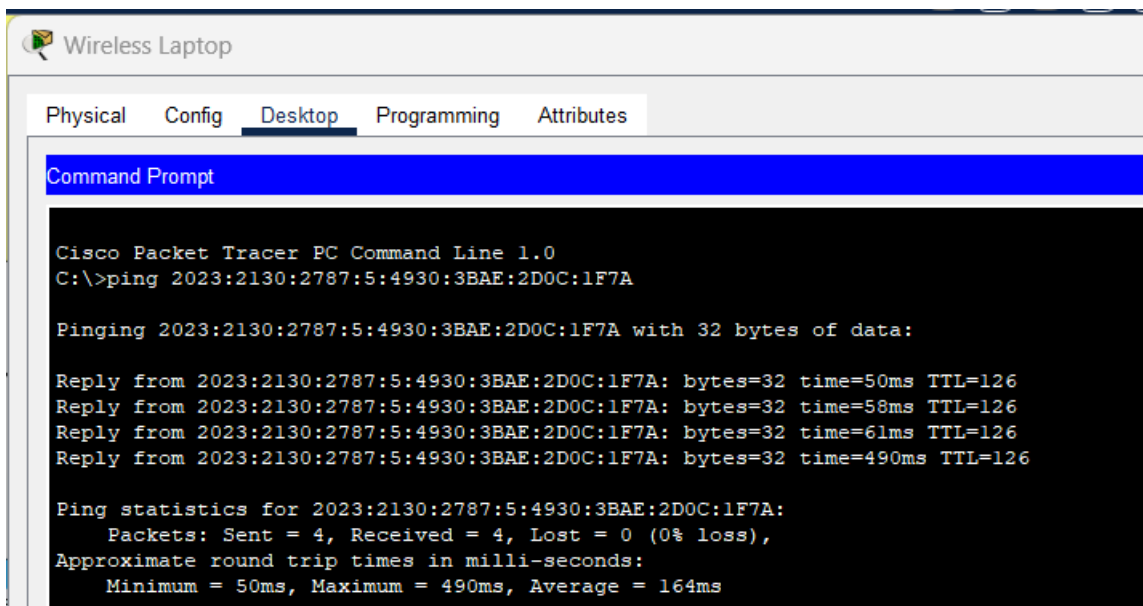


Figure 13.8: Shows interface for Serial1

14. CONNECTIVITY STATUS



```
Wireless Laptop
Physical Config Desktop Programming Attributes
Command Prompt

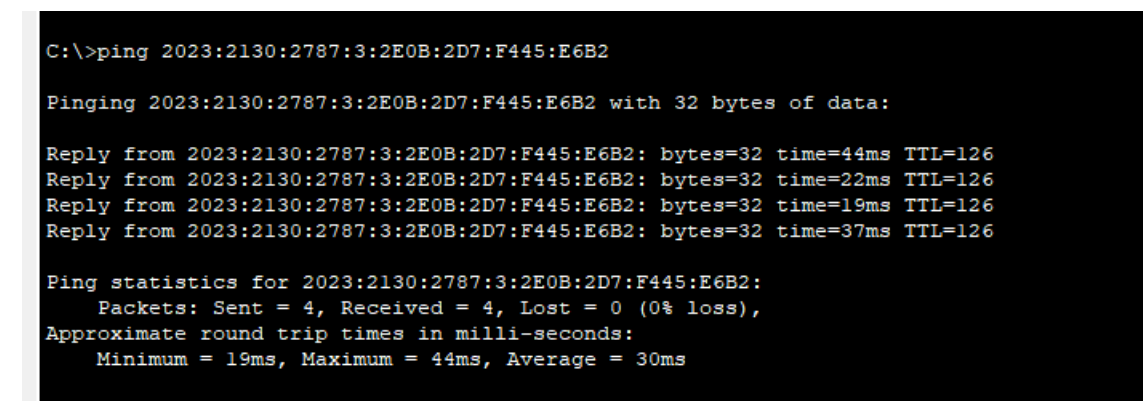
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 2023:2130:2787:5:4930:3BAE:2D0C:1F7A

Pinging 2023:2130:2787:5:4930:3BAE:2D0C:1F7A with 32 bytes of data:

Reply from 2023:2130:2787:5:4930:3BAE:2D0C:1F7A: bytes=32 time=50ms TTL=126
Reply from 2023:2130:2787:5:4930:3BAE:2D0C:1F7A: bytes=32 time=58ms TTL=126
Reply from 2023:2130:2787:5:4930:3BAE:2D0C:1F7A: bytes=32 time=61ms TTL=126
Reply from 2023:2130:2787:5:4930:3BAE:2D0C:1F7A: bytes=32 time=490ms TTL=126

Ping statistics for 2023:2130:2787:5:4930:3BAE:2D0C:1F7A:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 50ms, Maximum = 490ms, Average = 164ms
```

Figure 14.1. Ping from wireless laptop to LAB2_PC22



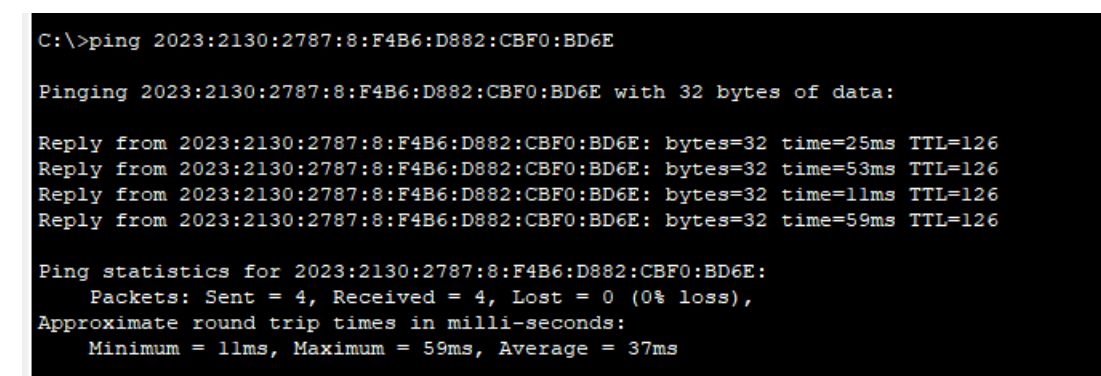
```
C:\>ping 2023:2130:2787:3:2E0B:2D7:F445:E6B2

Pinging 2023:2130:2787:3:2E0B:2D7:F445:E6B2 with 32 bytes of data:

Reply from 2023:2130:2787:3:2E0B:2D7:F445:E6B2: bytes=32 time=44ms TTL=126
Reply from 2023:2130:2787:3:2E0B:2D7:F445:E6B2: bytes=32 time=22ms TTL=126
Reply from 2023:2130:2787:3:2E0B:2D7:F445:E6B2: bytes=32 time=19ms TTL=126
Reply from 2023:2130:2787:3:2E0B:2D7:F445:E6B2: bytes=32 time=37ms TTL=126

Ping statistics for 2023:2130:2787:3:2E0B:2D7:F445:E6B2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 19ms, Maximum = 44ms, Average = 30ms
```

Figure 14.2 ping from wireless laptop to LAB1_PC22



```
C:\>ping 2023:2130:2787:8:F4B6:D882:CBF0:BD6E

Pinging 2023:2130:2787:8:F4B6:D882:CBF0:BD6E with 32 bytes of data:

Reply from 2023:2130:2787:8:F4B6:D882:CBF0:BD6E: bytes=32 time=25ms TTL=126
Reply from 2023:2130:2787:8:F4B6:D882:CBF0:BD6E: bytes=32 time=53ms TTL=126
Reply from 2023:2130:2787:8:F4B6:D882:CBF0:BD6E: bytes=32 time=11ms TTL=126
Reply from 2023:2130:2787:8:F4B6:D882:CBF0:BD6E: bytes=32 time=59ms TTL=126

Ping statistics for 2023:2130:2787:8:F4B6:D882:CBF0:BD6E:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 11ms, Maximum = 59ms, Average = 37ms
```

Figure 14.3 Ping from wireless laptop to ADMIN_2

```

Minimum = 11ms, Maximum = 59ms, Average = 37ms

C:\>ping 2023:2130:2787:7:5631:3A0D:2C7B:1EE8

Pinging 2023:2130:2787:7:5631:3A0D:2C7B:1EE8 with 32 bytes of data:

Request timed out.
Reply from 2023:2130:2787:7:5631:3A0D:2C7B:1EE8: bytes=32 time=20ms TTL=126
Reply from 2023:2130:2787:7:5631:3A0D:2C7B:1EE8: bytes=32 time=67ms TTL=126
Reply from 2023:2130:2787:7:5631:3A0D:2C7B:1EE8: bytes=32 time=44ms TTL=126

Ping statistics for 2023:2130:2787:7:5631:3A0D:2C7B:1EE8:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 20ms, Maximum = 67ms, Average = 43ms

```

Figure 14.4. Ping from wireless laptop to Sysadmin (management)

```

Minimum = 7ms, Maximum = 58ms, Average = 32ms

C:\>ping 2023:2130:2787:7:F6E1:DABD:BE89:A0F7

Pinging 2023:2130:2787:7:F6E1:DABD:BE89:A0F7 with 32 bytes of data:

Reply from 2023:2130:2787:7:F6E1:DABD:BE89:A0F7: bytes=32 time=25ms TTL=126
Reply from 2023:2130:2787:7:F6E1:DABD:BE89:A0F7: bytes=32 time=58ms TTL=126
Reply from 2023:2130:2787:7:F6E1:DABD:BE89:A0F7: bytes=32 time=40ms TTL=126
Reply from 2023:2130:2787:7:F6E1:DABD:BE89:A0F7: bytes=32 time=7ms TTL=126

Ping statistics for 2023:2130:2787:7:F6E1:DABD:BE89:A0F7:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 7ms, Maximum = 58ms, Average = 32ms

```

Figure 14.5. Ping wireless laptop to classroom PC4